

Manual

deeptrack[™]
by SPVH

Modbus Memory map

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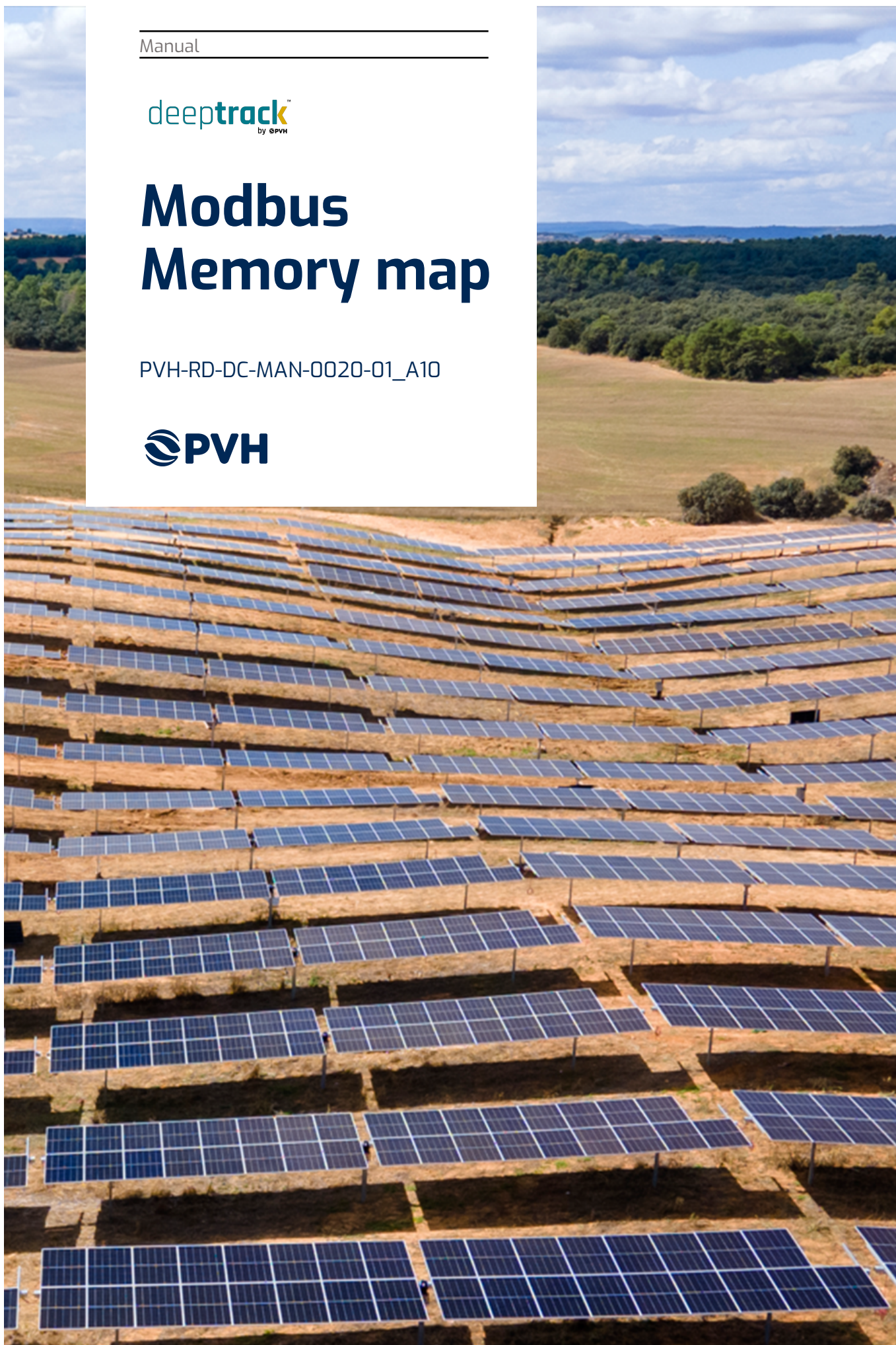


Table of Contents

- Glossary..... 3
- 1. Introduction..... 5
- 2. Memory map changelog..... 6
- 3. Modbus function code 9
- 4. Block numbering..... 10
- 5. Modbus client connection parameters 10
- 6. Server Port 502 - Modbus Memory Map 12
 - 6.1. Modbus map overview..... 12
 - 6.1.1. ID 255 - General TBox Map and Configuration..... 13
 - 6.1.2. ID 1 to 254 - Power-Block Map..... 13
 - 6.2. Modbus memory map registers..... 13
 - 6.2.1. Memory map distribution..... 13
 - 6.2.2. ID 255 - General TBox Map and Configuration..... 15
 - 6.2.3. ID 1 to 254 - Power Block Map 50
 - 6.2.4. Single Tracker Register Description..... 83
 - 6.3. Integration examples 91
 - 6.3.1. ID 255 - General TBox Map and Configuration..... 91
 - 6.3.2. ID 1 to 254 - Power-Block Map..... 92
 - 6.3.3. DBox status description 96
 - 6.3.4. Meteo status description..... 116
- 7. Server Port 503 - Modbus Memory Map 117
 - 7.1. Modbus map overview..... 117
 - 7.2. Modbus memory map registers 117
 - 7.2.1. Memory map distribution..... 117
 - 7.2.2. ID 1 to 254 - Power Block Map 118
 - 7.3. Integration examples 142
- 8. PVH Support..... 142

Glossary

Anemometer

A device used to measure wind speed and direction, providing critical data to monitor environmental conditions and optimize the performance, safety, and durability of the plant.

Backtracking

A stage of solar tracking during sunrise and sunset in which a specific control algorithm is applied to optimize energy production by adjusting the tracker position.

Block

A cluster of trackers connected to the same inverter.

DBox

PVH solution to control the tracker or manage the weather data communication.

Downlink

Communication from TBox to DBox, transmitting information such as setpoints, operation modes and alarms via the LoRaWAN protocol.

Emergency stop push button

Push-button situated on the DBox used to halt the tracker manually.

LoRa Gateway

Ethernet-LoRaWAN interface which allows communications between TBox and DBox.

Modbus

A communication protocol used in industrial automation systems for transmitting data between devices such as tracker controllers and sensors.

Sensor signal converter

Internal device used by PVH to read data from sensors.

Setpoint

The defined position of the tracker at a certain moment, calculated by TBox, to ensure optimal operation.

Stow position

A specific position of the solar tracker designed to safeguard its structural integrity during adverse weather conditions, such as high winds, snow or hail, and reduce the risk of damage to the tracker system.

TBox

Central controller of DeepTrack system. TBox is the solution developed by PVH to manage the tracker positions and the alarms according to the data collected by every DBox and sensor of the solar plant. It serves as the network control unit and establishes the plant's location as well as the time.

TPU

Device communicating the tracker position to the tracker control unit.

Tracker controller, Controller

Device which manages the motor movement of the solar tracker. Power cables come into the box to feed the motor.

Tracknet

PVH communication network solution for signal enhancement.

Uplink

Communication from each DBox to TBox, transmitting information such as communication status, actual position, setpoints, and alarms via the LoRaWAN protocol.

Weather station

A device, installed on the TBox towers, which measures atmospheric pressure, temperature, humidity, wind speed, and wind direction, transmitting the collected data to the Tracker manager cabinet for monitoring, analysis and alarm management.

1. Introduction

This document serves as a comprehensive guide to understanding and implementing the Modbus protocol for seamless integration between Supervisory Control and Data Acquisition (SCADA) systems and photovoltaic plant equipment. It provides an in-depth understanding of the Modbus memory map structure and its application in monitoring and controlling PV plant systems.

The Modbus protocol is a widely adopted standard in industrial automation and energy management systems, known for its simplicity, reliability, and scalability. In PV plants, it plays a critical role in enabling real-time monitoring and control of key components such as weather stations and trackers.

This manual focuses on the Modbus memory map, a structured layout of registers that allows SCADA systems to retrieve and command critical data points for real-time operation, diagnostics, and performance optimization.

The manual is structured to address the unique needs of PV plant operations. For instance, **Server Port 502** provides access to **general plant information** (control states, information of DBox and other network devices) **and tracker commands**, while **Server Port 503** focuses on **TPU** (Tracker Position Unit) **data**.

This manual includes:

- A **TBox version changelog stating the Modbus map changes** introduced on each TBox version.
- **Modbus Map Overview:** A detailed explanation of how the memory map is organized, including the division of registers into logical sections such as general plant information, power block data, and device-specific registers.
- **Memory Map Distribution:** A summary of register locations, grouped by device type and functionality for easy reference.
- **Register Descriptions:** Comprehensive tables detailing the purpose, behavior, and configuration of each register, including scaling factors and data types.
- **Integration Examples:** Practical use cases illustrating how to command devices and interpret data, such as setting tracker positions or understanding weather station outputs.
- **Version-Specific Details:** Information on memory map changes introduced in different firmware versions, ensuring compatibility with TBox and DBox.

For each server port, slave id 255 is reserved for general plant information and slave ids 1 to 254 are reserved for block grouped devices. PVH system control devices are logically divided into groups called blocks.

The server port 502 contains differences for system versions 3.1 and 3.2. This is given by the DBox firmware version deployed at the PV plant. Detailed information can be found in the integration examples.

This document aims to provide an in-depth exploration of the innovative solution developed by PV Hardware Solutions (PVH) for optimizing solar plant operations. It highlights the system's technological advancements, implementation strategies and operational benefits. The solution encompasses state-of-the-art algorithms, robust hardware components, an efficient communication architecture and an integrated control system to ensure seamless operation and maintenance of solar plants.

2. Memory map changelog

The following table outlines the Modbus memory map changes introduced in different Modbus server versions and their corresponding TBox versions. This information helps SCADA integrators understand the available Modbus registers for their specific TBox version.

TBox VERSION	MODBUS SERVER VERSION	MEMORY MAP CHANGELOG
v3.2.19.0	v4.6.0	Added <i>WindSpeedEffective</i>. See registers in ID 255 - General TBox Map and Configuration (► 15) and ID 1 to 254 - Power Block Map (► 50) for details.
v3.2.18.3 v5.0.5.3	v4.4.5	Added <i>FuotaState</i>. See register 40038 in ID 255 - General TBox Map and Configuration (► 15) (Global Status) for details.
v5.0.3.0.1	v4.3.0	Internal improvements
v3.2.18.1.0 - v3.2.18.2.0	v4.2.0	- Introduced Hail Algorithm. See registers 40033–40037 in ID 255 - General TBox Map and Configuration (► 15) (Global Status) and registers 40166–40167 (Automode). - Added <i>SnowForce</i>, <i>SnowForceReturnCode</i>, and <i>WindForceReturnCode</i>. See registers 40168–40170 in ID 255 - General TBox Map and Configuration (► 15) (Automode).
v3.2.17.1.0	v4.0.7	Internal improvements
v3.2.14.4 - v3.2.15.0.0	v4.0.6	Internal improvements
v3.2.14.1 - v3.2.14.3.1	v4.0.5	Internal improvements
v3.2.13.1	v4.0.3	- Included multiple Meteo data per block. See registers 48275 to 48424 of section ID 1 to 254 - Power Block Map (► 50) for more information. - Included Sensor Signal Converter sensors (wind, air, snow and hail) data. See registers 48425 to 48784 of section ID 1 to 254 - Power Block Map (► 50) for more information.
v3.2.12.4 - v3.2.12.5	v3.0.7	Internal improvements.
v3.2.12.3	v3.0.6	Internal improvements.
v3.2.12.1 - v3.2.12.2	v3.0.5	Internal improvements.

TBox VERSION	MODBUS SERVER VERSION	MEMORY MAP CHANGELOG
v3.2.12.0	v3.0.3	Included Flood data. See registers 40300 to 40459 of section ID 255 - General TBox Map and Configuration (► 15) for more information.
v3.2.11.0 - v3.2.11.5	v3.0.0	Modbus server permits renumbering the blocks. See section Block numbering (► 10) for more information.
v3.2.10.1 - v3.2.10.3	v2.8.0	Included registers <i>SetpointLowerLimit</i> , <i>SetpointUpperLimit</i> , <i>SetpointLimitEnable</i> and <i>SetpointLimitsInternalEnabled</i> in Automode section. See registers 40162 to 40165 of section ID 255 - General TBox Map and Configuration (► 15) for more information.
v3.2.9.7	v2.7.1	Internal improvements.
v3.2.8.9 - v3.2.9.6	v2.7.0	Included registers <i>SnowUnlockable</i> and <i>SnowUnlock</i> in Automode section, on registers 40160 and 40161 respectively. See ID 255 - General TBox Map and Configuration (► 15) for more information.
v3.2.8.6 - v3.2.8.8	v2.6.5	Internal improvements.
v3.2.8.2 - v3.2.8.5	v2.6.4	Internal improvements.
v3.2.8.1	v2.6.1	- Included <i>WindForce</i> in Automode section. See register 40159 of section ID 255 - General TBox Map and Configuration (► 15) for more information. - Included registers <i>SnowIsEnabled</i>, <i>DiffuselsEnabled</i> and <i>HailsEnabled</i> in Automode section, on registers 40152, 40156 and 40158 respectively. See section ID 255 - General TBox Map and Configuration (► 15) for more information.
		Deprecated registers <i>SnowEnable</i>, <i>ForceSnowCompleteMovementRangeTrigger</i>, <i>ForceSnowLimitedMovementRangeTrigger</i>, <i>DiffuseEnable</i> and <i>HailEnable</i> in Automode section, on registers 40151, 40153, 40154, 40155 and 40157. See section ID 255 - General TBox Map and Configuration (► 15) for more information.
		Deprecated Input Data section. See

TBox VERSION	MODBUS SERVER VERSION	MEMORY MAP CHANGELOG
v3.2.8.1	v2.6.1	registers 40200 to 40259 of section ID 255 - General TBox Map and Configuration (► 15) for more information.
v3.1.7.21 - v3.2.7.28	v2.5.0	Integrate Tracknet data. See registers 48175 to 48274 of section ID 1 to 254 - Power Block Map (► 50) for more information.
v3.1.7.10 - v3.1.7.20	v2.4.0	• Included Gateway data. See registers 48050 to 48174 of section ID 1 to 254 - Power Block Map (► 50) for more information. • Included register <i>Notifications</i> in Meteo 0 device. See register 40028 of section ID 1 to 254 - Power Block Map (► 50) for more information. • Extended number of DBox from 250 to 500 per block. See registers 40050 to 48049 of section ID 1 to 254 - Power Block Map (► 50) for more information.
v3.1.7.1 - v3.1.7.9	v2.3.0	First version of the Modbus memory map

3. Modbus function code

Modbus function codes are a cornerstone of the Modbus protocol, defining the specific operations that a Modbus client can execute on a remote server or device. Each function code corresponds to a distinct command, enabling a wide range of actions such as reading data, writing commands, or diagnosing communication and device conditions.

CODE	NAME	DESCRIPTION
01	Read Coils	This function code is used to read from 1 to 2000 contiguous status of coils in a remote device.
02	Read discrete inputs	This function code is used to read from 1 to 2000 contiguous status of discrete inputs in a remote device.
03	Read holding registers	This function code is used to read the contents of a contiguous block of holding registers in a remote device.
04	Read input registers	This function code is used to read from 1 to 125 contiguous input registers in a remote device.
05	Write single coil	This function code is used to write a single output to either ON or OFF in a remote device.
06	Write single register	This function code is used to write a single holding register in a remote device.
07	Read exception status	This function code is used to read the contents of 8 Exception Status outputs in a remote device.
08	Diagnostics	Modbus function code 08 provides a series of tests for checking the communication system between a client device and a server, or for checking various internal error conditions within a server.
15	Write multiple coils	This function code is used to force each coil in a sequence of coils to either ON or OFF in a remote device.
16	Write multiple registers	This function code is used to write a block of contiguous registers, 1 to 123, in a remote device.
17	Report server ID	This function code is used to read the description of the type, the current status, and other information specific in a remote device.
20	Read file record	This function code is used to perform a file record read.
21	Write file record	This function code is used to perform a file record write.
22	Mask write register	This function code is used to modify the contents of a specified holding register using a combination of an AND mask, an OR mask, and the register's current contents. The function can be used to set or clear individual bits in the register.
23	Read/Write Multiple registers	This function code performs a combination of one read operation, and one write operation in a single Modbus transaction.
24	Read FIFO Queue	This function code allows reading the contents of a First-In-First-Out (FIFO) queue of registers in a remote device.

4. Block numbering

NOTICE



Block numbering configuration

This functionality is configured internally and must be requested from the commissioning team.

The Modbus server allows renumbering blocks for large plants. When the block numbers in one TBox exceed 254, they can be represented on a Modbus server by renumbering the lower block numbers starting from one and applying the same operation to every subsequent block.

For example, if one TBox manages blocks 301, 302, and 305, it can be configured to address block 301 on slave ID 1, block 302 on slave ID 2, and block 305 on slave ID 5.

Similarly, if one TBox manages blocks 55, 57, and 60, block 55 is addressed on slave ID 1, block 57 on slave ID 3, and block 60 on slave ID 6.

The SCADA interface is responsible for registering the original block IDs, if necessary.

5. Modbus client connection parameters

This chapter presents a summary table of recommended operational parameters to be configured on Modbus clients. By following them, the Modbus server shall be able to response successfully to every request.

The recommended connection parameters are:

- **Span gaps** parameter is **enabled**, so that non-consecutive registers can be read in the same request.
- **Request size** corresponds to 80 registers.

The generated table was calculated based on the following context parameters:

- **Single request time** corresponds to 8 milliseconds, measured in laboratory conditions and **for one Modbus client**.
- Every Modbus client consults the same number of Modbus registries and applies the same configuration.

Table 1. Parameters values for defined context parameters estimation

CONCURRENT REQUESTS OR SIMULTANEOUS CLIENTS	NUMBER OF REQUESTED REGISTERS PER CLIENT	ESTIMATED MINIMUM SAMPLING PERIOD/MS
1	10 000	1000
1	20 000	2000
1	30 000	3000
1	40 000	4000
2	10 000	4000
2	20 000	8000
2	30 000	12 000
2	40 000	16 000
3	10 000	9000
3	20 000	18 000
3	30 000	27 000
3	40 000	36 000
4	10 000	16 000
4	20 000	32 000
4	30 000	48 000
4	40 000	64 000

Analytically, this is the applicable equation to calculate the estimated minimum sampling period:

$$NumberOfRequests = \sum_{PerModbusClient} \frac{TotalNumberOfRegistries}{RequestSize}$$

MinimumSamplingPeriod = SingleRequestTime × NumberOfRequests × ConcurrentRequests

The above equations must be used if a single request time exceeds 8 milliseconds or if any context parameter is not met.

For example, assuming one of the previous table cases and the following context parameters:

- Concurrent Requests = 3 clients
- NumberOfRequestedRegistersPerClient = 20 000 registers/client
- SingleRequestTime = 8 milliseconds
- RequestSize = 80 registers

Thus, the NumberOfRequests is calculated as stated in the following example:

$$\begin{aligned}
 NumberOfRequests &= \left(\frac{20\,000\,registers}{80\,registers} \right)_{client\,1} + \left(\frac{20\,000\,registers}{80\,registers} \right)_{client\,2} \\
 &+ \left(\frac{20\,000\,registers}{80\,registers} \right)_{client\,3} = 750\,requests
 \end{aligned}$$

Figure 1. NumberOfRequests calculation example

Thus, the MinimumSamplingPeriod is calculated as stated in the following example:

MinimumSamplingPeriod = 8 milliseconds × 750 requests × 3 clients = 18 000 milliseconds

As stated in the table, the equations can be extended to support multiple Modbus clients operating with different connection parameters.

Network traffic and Modbus client computing resources will affect the single request time. It is important to account for a margin in the configured sampling period to handle transient dynamics in the network and application computing resources.

Operational parameters must be recalculated if any context conditions change, such as the number of connected clients, the number of requested registers, or the network traffic.

6. Server Port 502 - Modbus Memory Map

6.1. Modbus map overview

This section describes the information found within Modbus port 502. Connect a Modbus client to this port to retrieve the following information.

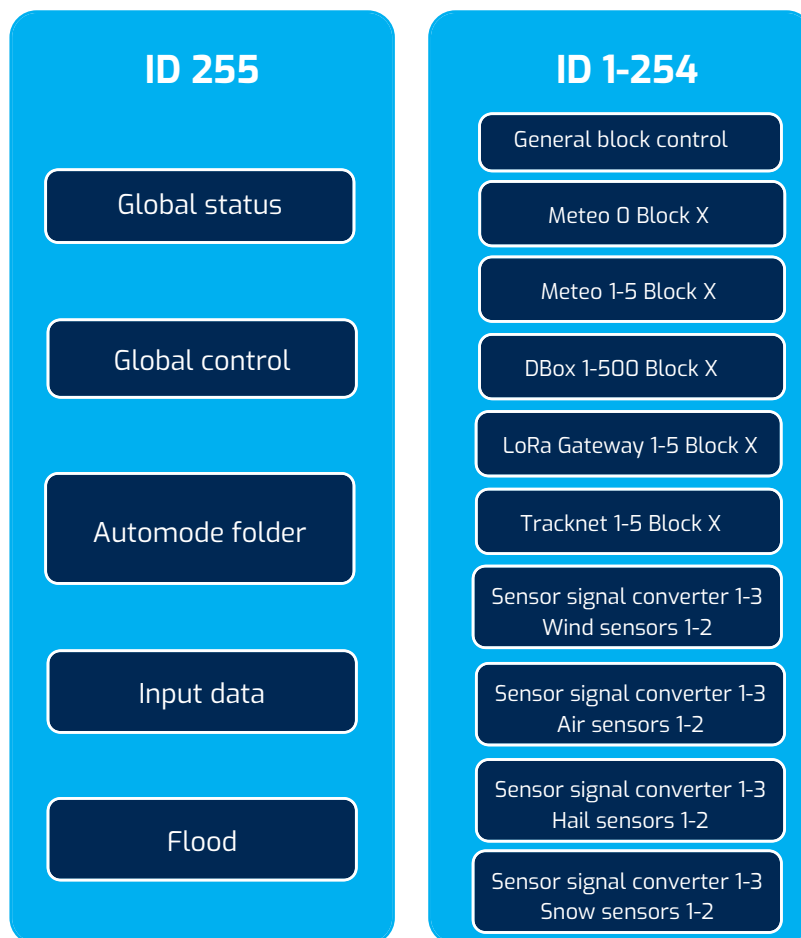


Figure 2. Modbus memory map overview

6.1.1. ID 255 - General TBox Map and Configuration

TBox is designed to simplify SCADA readings through the structure of its Modbus map.

Its architecture allows the following

- **Global status:** Read the plant information, TBox operation mode and active alarms, as well as the wind information and UPS status.
- **Global control:** Write the parameters that enable SCADA to control the whole plant.
- **Automode:** Write and read the set configuration within Auto operation-mode. It allows to enable or disable the managements of snow, diffuse and hail.
- **Flood:** Read the information of each Flood zone. This section provides information of the setpoint limits of the zone and its associated flood sensor value.

6.1.2. ID 1 to 254 - Power-Block Map

The number of the desired block to be connected corresponds to the Modbus ID used for the connection. Each block is configured to display one Meteo and up to 500 DBox entries. The DBox entries are ordered consecutively.

- **General Block Control:** As the connection ID simulates the block, it is allowed to control a whole block.
- **Meteo X Block X:** Commonly, every block has its own Meteo, so the connected ID simulates the block, and it displays all the Meteo information if installed.
- **DBox Block X:** Inside the desired Block (simulated by the connected ID) it is possible to control the selected DBox or read its information. Read and write parameters where SCADA can control a single DBox.
- **LoRa Gateway Block X:** Permits monitoring the state of LoRa Gateways.
- **Tracknet Block X:** Permits monitoring the state of LoRa Tracknet devices.
- **Sensor Signal Converter Block X:** Read information of wired sensors: wind, hail, snow and atmosphere.

6.2. Modbus memory map registers

6.2.1. Memory map distribution

TAG	MODBUS ID	FIRST ADDRESS	LAST ADDRESS	DESCRIPTION
GLOBAL STATUS FOLDER	255	40000	40099	-
GLOBAL CONTROL FOLDER	255	40100	40149	-
AUTOMODE FOLDER	255	40150	40199	PLANT CONTROL
INPUT DATA	1-254	40200	40249	DEPRECATED
FLOOD	1-254	40300	40459	-
GENERAL BLOCK CONTROL	1-254	40000	40019	DBOX BLOCK CONTROL

TAG	MODBUS ID	FIRST ADDRESS	LAST ADDRESS	DESCRIPTION
METEO 0	1-254	40020	40049	METEO OR INFORMATION WITHIN POWER BLOCK
DBOX 1-500	1-254	40050	48049	DBOX INFORMATION AND CONTROL WITHIN POWER BLOCK
GATEWAY 1-5	1-254	48050	48174	GATEWAY INFORMATION WITHIN POWER BLOCK
TRACKNET 1-5	1-254	48175	48274	TRACKNET INFORMATION WITHIN POWER BLOCK
METEO 1-5	1-254	48275	48424	METEO 1-5 INFORMATION WITHIN POWER BLOCK
SENSOR SIGNAL CONVERTERS 1-3 WIND SENSORS 1-2	1-254	48425	48604	UP TO 3 CONVERTERS PER BLOCK AND 2 SENSORS PER CONVERTER
SENSOR SIGNAL CONVERTERS 1-3 AIR SENSORS 1-2	1-254	48605	48664	UP TO 3 CONVERTERS PER BLOCK AND 2 SENSORS PER CONVERTER
SENSOR SIGNAL CONVERTERS 1-3 HAIL SENSORS 1-2	1-254	48665	48724	UP TO 3 CONVERTERS PER BLOCK AND 2 SENSORS PER CONVERTER
SENSOR SIGNAL CONVERTERS 1-3 SNOW SENSORS 1-2	1-254	48725	48784	UP TO 3 CONVERTERS PER BLOCK AND 2 SENSORS PER CONVERTER

6.2.2. ID 255 - General TBox Map and Configuration

GLOBAL STATUS										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	YearUTC	40000	0	Uint	YYYY	0	65 535	1	R	Local year
FEEDBACK	MonthUTC	40001	1	Uint	MM	1	12	1	R	Local month
FEEDBACK	DayUTC	40002	2	Uint	DD	1	31	1	R	Local day
FEEDBACK	HourUTC	40003	3	Uint	hh	0	23	1	R	Local 24 hour
FEEDBACK	MinuteUTC	40004	4	Uint	mm	0	59	1	R	Local minute
FEEDBACK	SecondUTC	40005	5	Uint	ss	0	59	1	R	Local second
FEEDBACK	SunAzimuth	40006	6	Uint	°	0	36 000	100	R	Sun Azimuth
FEEDBACK	SunZenith	40007	7	Int	°	-18 000	18 000	100	R	Sun Zenith
FEEDBACK	SunAngle	40008	8	Int	°	-18 000	18 000	100	R	Sun Angle
FEEDBACK	GlobalAutoElevationTarget	40009	9	Int	°	-7000	7000	100	R	Sun tracking setpoint (degrees from horizontal) for trackers in auto mode
FEEDBACK	GlobalWindDirectionEW	40010	10	Uint	-	-	-	-	R	Stow Control Mode: 1-East 2-West
FEEDBACK	BacktrackingStatus	40011	11	Uint	-	0	1	-	R	Status of Backtracking
FEEDBACK	NightSetpoint	40012	12	Int	°	-7000	7000	100	R	Setpoint defined for the Night position
FEEDBACK	NightMode	40013	13	Uint	-	0	1	-	R	Status of Night Mode

GLOBAL STATUS										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	WindDirection	40014	14	Uint	°	0	360	1	R	Numeric value of wind direction for TBox 3.1.
FEEDBACK	WindSpeed	40015	15	Uint	m/s	0	4500	100	R	Numeric value of wind speed for TBox 3.1
FEEDBACK	WindAlarm	40016	16	Uint	-	0	1	-	R	Status of Wind Speed Alarm
FEEDBACK	WindInactivity Alarm	40017	17	Uint	-	0	1	-	R	If Wind_Speed is 0 for more than 24 hours, WindInactivityAlarm gets TRUE. Check the Anemometer
FEEDBACK	UpsAlarm	40018	18	Uint	-	0	1	-	R	UPS Alarm Status. If UpsAlarm is TRUE, it is in alarm. Check the UPS
FEEDBACK	UpsBattery	40019	19	Uint	-	0	1	-	R	If UpsBattery is TRUE, UPS is the TBox power supply
FEEDBACK	UpsCharging	40020	20	Uint	-	0	1	-	R	If UpsCharging is TRUE, Auxiliary Services is the TBox power supply
FEEDBACK	Anemometer ActiveNumber	40021	21	Uint	-	0	65 535	1	R	Count of active anemometers considering all Meteo and TBox anemometers
FEEDBACK	AnemometerTotalNumber	40022	22	Uint	-	0	65 535	1	R	Count of total anemometers considering all Meteo and TBox anemometers

GLOBAL STATUS										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Anemometer Warning	40023	23	Uint	-	0	1	-	R	Warnings are raised when at least one anemometer is not communicating with TBox considering all Meteo and TBox anemometers
FEEDBACK	SnowState	40024	24	Uint	-	0	65 535	-	R	Snow Control State Machine state
FEEDBACK	InternalSnowSensorValue	40025	25	Uint	cm	0	1000	1	R	Snow internal sensor value
FEEDBACK	ApplicableSnowSensorValue	40026	26	Uint	cm	0	1000	1	R	Snow applicable sensor value
FEEDBACK	DiffuseState	40027	27	Uint	-	0	65 535	-	R	Diffuse Control State Machine state
FEEDBACK	InternalDiffuseSensorValue	40028	28	Uint	%	0	10 000	100	R	Diffuse-Global irradiance ratio internal value
FEEDBACK	ApplicableDiffuseSensorValue	40029	29	Uint	%	0	10 000	100	R	Diffuse-Global irradiance ratio applicable value
Hail Control State Machine state:										
FEEDBACK	HailState	40030	30	Uint	-	0	65 535	-	R	0 - Not alarmed 1 - Alarmed 2 - Alarm counting down
FEEDBACK	InternalHailSensorValue	40031	31	Uint	-	0	1	-	R	Hail internal sensor value

GLOBAL STATUS										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	ApplicableHail SensorValue	40032	32	Uint	-	0	1	-	R	Hail applicable sensor value
										0 - no alarm
FEEDBACK	PredictiveHail Alarm	40033	33	Uint	-	0	32767	-	R	1 - hail alarm
										32767 - no data
FEEDBACK	PredictiveHear beat	40034	34	Uint	-	0	1	-	R	0 - Receiving predicted hail data
										1 - Missing predicted hail data (1h)
FEEDBACK	PredictiveData ValidTimeout	40035	35	Uint	-	0	1	-	R	0 - The time frame of the predicted hail data is currently valid.
										1 - Outdated predicted hail data.
										0 - hail alarm disabled
FEEDBACK	HailingDataSo urce	40036	36	Uint	-	0	255	-	R	1 - predictive hail
										2 - sensor hail
										128 - no data source
FEEDBACK	HailStowDirec tion	40037	37	Uint	-	1	2	-	R	1 - East
										2 - West
FEEDBACK	FuotaState	40038	38	Uint	-	0	1	-	R	0 - Inactive
										1 - Active

GLOBAL CONTROL										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
CONTROL	GlobalMessageCounter	40100	100	Uint	-	0	65 535	-	RW	Auto incremented counter when a message is processed by TBox It does not mean that the message is sent to or received by DBox since LoRa communications are not immediate. Status or setpoint feedback will be available once DBox sends its state.
CONTROL	WindAlarmTrigger	40101	101	Uint	-	0	1	-	RW	Signal to send a global wind alarm to all trackers
CONTROL	StopAlarmTrigger	40102	102	Uint	-	0	1	-	RW	Signal to send a global stop alarm to all trackers
CONTROL	ManualElevationTargetTrigger	40103	103	Uint	-	0	1	-	RW	Signal to send a global manual setpoint to all trackers
CONTROL	ManualElevationTarget	40104	104	Int	°	-7000	7000	100	RW	Global manual setpoint to be sent
CONTROL	SetAutoControlTrigger	40105	105	Uint	-	0	1	-	RW	Signal to send all trackers a set to auto signal
CONTROL	ConstructionTrigger	40106	106	Uint	-	0	1	-	RW	Signal to send all trackers a set to construction position signal

AUTOMODE										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	TriggerCounter	40150	150	Uint	-	-	-	-	RW	Feedback from Trigger messages
CONTROL	SnowEnable	40151	151	Uint	-	0	1	-	RW	Deprecated
FEEDBACK	SnowIsEnabled	40152	152	Uint	-	0	1	-	R	Feedback from Snow Enable
CONTROL	ForceSnowCompleteMovementRangeTrigger	40153	153	Uint	-	0	1	-	RW	Deprecated
CONTROL	ForceSnowLimitedMovementRangeTrigger	40154	154	Uint	-	0	1	-	RW	Deprecated
CONTROL	DiffuseEnable	40155	155	Uint	-	0	1	-	RW	Deprecated
FEEDBACK	DiffuseIsEnabled	40156	156	Uint	-	0	1	-	R	Feedback from Diffuse Enable
CONTROL	HailEnable	40157	157	Uint	-	0	1	-	RW	Deprecated
FEEDBACK	HailIsEnabled	40158	158	Uint	-	0	1	-	R	Feedback from Hail Enable
Value 0: No force										
CONTROL	WindForce	40159	159	Uint	-	0	2	-	R	Value 1: Force wind stow position to east Value 2: Force wind stow position to west

AUTOMODE										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	SnowUnlockable	40160	160	Uint	-	0	1	-	R	Value 0: The user cannot manually unlock the plant from snow control Value 1: The user can manually unlock the plant from snow control
CONTROL	SnowUnlock	40161	161	Uint	-	0	1	-	RW	Value 0: No action Value 1: The user sends the instructions to the plant from snow control
CONTROL	SetpointLower Limit	40162	162	Uint	°	-90	90	-	RW	If the SetpointLimitsInternalEnabled variable is set to 1, it allows the user to set the lower limit of the setpoint. This value must be lower than the upper limit and must be within the project's movement range (set in commissioning).
CONTROL	SetpointUpper Limit	40163	163	Uint	°	-90	90	-	RW	If the SetpointLimitsInternalEnabled variable is set to 1, it allows the user to set the lower limit of the setpoint.

AUTOMODE										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
CONTROL	SetpointUpper Limit	40163	163	Uint	°	-90	90	-	RW	This value must be lower than the upper limit and must be within the project's movement range (set in commissioning). If the SetpointLimitsInternalEnabled variable is set to 1, it allows the user to set the lower limit of the setpoint.
CONTROL	SetpointLimitEnabled	40164	164	Uint	-	0	1	-	RW	Value 0: Setpoint limits are not applied, and tracking is performed with default project's limits. Value 1: Setpoint limits are applied to commands sent from TBox. This value is set by the SCADA user. NOTICE: This functionality is not compatible with Backtracking 3D functionality.
FEEDBACK	SetpointLimitsInternalEnabled	40165	165	Uint	-	0	1	-	R	Value 0: Modifying the setpoint limits of the plant is not permitted. Value 1: Allows the OPC-UA interface to limit setpoint values

AUTOMODE										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	SetpointLimits InternalEnabled	40165	165	Uint	-	0	1	-	R	This value is set by the project commissioner or the aftersales team. Contact them to activate this functionality
0: No force										
CONTROL	HailForce	40166	166	Uint	-	0	2	-	RW	1: Force hail stow position to east 2: Force hail stow position to west
0 - forced command valid and applied										
1 - internal error										
FEEDBACK	HailForceReturnCode	40167	167	Uint	-	0	255	-	R	129 - Invalid forced command because requested direction is not compatible with the current hail stow position since wind is directional
Other values are reserved for future usage										

AUTOMODE										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
CONTROL	SnowForce	40168	168	Uint	-	0	2	-	RW	0: No force
										1: Force snow stow position to east
										2: Force snow stow position to west
FEEDBACK	SnowForceReturnCode	40169	169	Uint	-	0	255	-	R	0 - forced command valid and applied
										1 - internal error
										129 - Invalid forced command because requested direction is not compatible with the current snow stow position since wind is directional
FEEDBACK	WindForceReturnCode	40170	170	Uint	-	0	255	-	R	0 - forced command valid and applied
										1 - internal error
										129 - Invalid forced command because requested direction is not compatible with the current wind stow position since wind is directional

INPUT DATA										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
CONTROL	SnowDataFromSCADA	40200	200	Uint	cm	0	1000	1	RW	Deprecated
CONTROL	SelectSCADASnowData	40201	201	Uint	-	0	1	-	RW	Deprecated
FEEDBACK	SCADASnowDataIsSelected	40202	202	Uint	-	0	1	-	R	Deprecated
CONTROL	GlobalIrradianceFromSCADA	40203	203	Uint	W/m ²	0	20 000	10	RW	Deprecated
CONTROL	DiffuseIrradianceFromSCADA	40204	204	Uint	W/m ²	0	20 000	10	RW	Deprecated
CONTROL	SelectSCADADiffuseData	40205	205	Uint	-	0	1	-	RW	Deprecated
FEEDBACK	SCADADiffuseDataIsSelected	40206	206	Uint	-	0	1	-	R	Deprecated
CONTROL	HailSensorFromSCADA	40207	207	Int	-	0	1	-	RW	Deprecated
CONTROL	SelectSCADAHailData	40208	208	Uint	-	0	1	-	RW	Deprecated
FEEDBACK	SCADAHailDataIsSelected	40209	209	Uint	-	0	1	-	R	Deprecated

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone0.AngleLimit	40300	300	Uint	°	0	255	1	R	Maximum angle movement permitted to trackers in automatic mode. This value can be overwritten with a manual setpoint. Value 254 means that DBox in this flood zone is applying its latest received angle limit. Value 255 means that DBox in this flood zone is not applying any angle limit.
FEEDBACK	Zone0.Sensor Code	40301	301	Uint	-	0	3	-	R	Sensor code explaining the quality of the input data. - Value 0 stands for "OK" - Value 1 stands for "No sensor data" - Value 2 stands for "Wrong sensor data" - Value 3 stands for "Waiting for sensor data, applying last angle limit"
FEEDBACK	Zone0.FloodLimitActive	40302	302	Uint	-	0	1	-	R	Flood limit active states if any movement limit is being applied due to flooding sensor readings. Value 0: "Inactive" Value 1: "Active"

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone0.FloodZoneEnabled	40303	303	Uint	-	0	1	-	R	Flood zone enabled states if this flood zone is enabled for limiting angle movement due to flooding sensor readings. Value 0: "Disabled" Value 1: "Enabled"
FEEDBACK	Zone0.FloodLevel	40304	304	Uint	m	0	255	100	R	Measured height of water from the ground.
FEEDBACK	Zone1.AngleLimit	40310	310	Uint	°	0	255	1	R	Maximum angle movement permitted to trackers in automatic mode. This value can be overwritten with a manual setpoint. Value 254 means that DBox in this flood zone is applying its latest received angle limit. Value 255 means that DBox in this flood zone is not applying any angle limit.

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone1.SensorCode	40311	311	Uint	-	0	3	-	R	Sensor code explaining the quality of the input data. - Value 0 stands for "OK" - Value 1 stands for "No sensor data" - Value 2 stands for "Wrong sensor data" - Value 3 stands for "Waiting for sensor data, applying last angle limit"
FEEDBACK	Zone1.FloodLimitActive	40312	312	Uint	-	0	1	-	R	Flood limit active states if any movement limit is being applied due to flooding sensor readings. Value 0: "Inactive" Value 1: "Active"
FEEDBACK	Zone1.FloodZoneEnabled	40313	313	Uint	-	0	1	-	R	Flood zone enabled states if this flood zone is enabled for limiting angle movement due to flooding sensor readings. Value 0: "Disabled" Value 1: "Enabled"
FEEDBACK	Zone1.FloodLevel	40314	314	Uint	m	0	255	100	R	Measured height of water from the ground.

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone2.AngleLimit	40320	320	Uint	°	0	255	1	R	Maximum angle movement permitted to trackers in automatic mode. This value can be overwritten with a manual setpoint. Value 254 means that DBox in this flood zone is applying its latest received angle limit. Value 255 means that DBox in this flood zone is not applying any angle limit.
FEEDBACK	Zone2.Sensor Code	40321	321	Uint	-	0	3	-	R	Sensor code explaining the quality of the input data. - Value 0 stands for "OK" - Value 1 stands for "No sensor data" - Value 2 stands for "Wrong sensor data" - Value 3 stands for "Waiting for sensor data, applying last angle limit"
FEEDBACK	Zone2.FloodLimitActive	40322	322	Uint	-	0	1	-	R	Flood limit active states if any movement limit is being applied due to flooding sensor readings. Value 0: "Inactive" Value 1: "Active"

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone2.FloodZoneEnabled	40323	323	Uint	-	0	1	-	R	Flood zone enabled states if this flood zone is enabled for limiting angle movement due to flooding sensor readings. Value 0: "Disabled" Value 1: "Enabled"
FEEDBACK	Zone2.FloodLevel	40324	324	Uint	m	0	255	100	R	Measured height of water from the ground.
FEEDBACK	Zone3.AngleLimit	40330	330	Uint	°	0	255	1	R	Maximum angle movement permitted to trackers in automatic mode. This value can be overwritten with a manual setpoint. Value 254 means that DBox in this flood zone is applying its latest received angle limit. Value 255 means that DBox in this flood zone is not applying any angle limit.

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone3.Sensor Code	40331	331	Uint	-	0	3	-	R	Sensor code explaining the quality of the input data. - Value 0 stands for "OK" - Value 1 stands for "No sensor data" - Value 2 stands for "Wrong sensor data" - Value 3 stands for "Waiting for sensor data, applying last angle limit"
FEEDBACK	Zone3.FloodLimitActive	40332	332	Uint	-	0	1	-	R	Flood limit active states if any movement limit is being applied due to flooding sensor readings. Value 0: "Inactive" Value 1: "Active"
FEEDBACK	Zone3.FloodZoneEnabled	40333	333	Uint	-	0	1	-	R	Flood zone enabled states if this flood zone is enabled for limiting angle movement due to flooding sensor readings. Value 0: "Disabled" Value 1: "Enabled"
FEEDBACK	Zone3.FloodLevel	40334	334	Uint	m	0	255	100	R	Measured height of water from the ground.

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone4.AngleLimit	40340	340	Uint	°	0	255	1	R	Maximum angle movement permitted to trackers in automatic mode. This value can be overwritten with a manual setpoint. Value 254 means that DBox in this flood zone is applying its latest received angle limit. Value 255 means that DBox in this flood zone is not applying any angle limit.
FEEDBACK	Zone4.Sensor Code	40341	341	Uint	-	0	3	-	R	Sensor code explaining the quality of the input data. - Value 0 stands for "OK" - Value 1 stands for "No sensor data" - Value 2 stands for "Wrong sensor data" - Value 3 stands for "Waiting for sensor data, applying last angle limit"
FEEDBACK	Zone4.FloodLimitActive	40342	342	Uint	-	0	1	-	R	Flood limit active states if any movement limit is being applied due to flooding sensor readings. Value 0: "Inactive" Value 1: "Active"

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone4.FloodZoneEnabled	40343	343	Uint	-	0	1	-	R	Flood zone enabled states if this flood zone is enabled for limiting angle movement due to flooding sensor readings. Value 0: "Disabled" Value 1: "Enabled"
FEEDBACK	Zone4.FloodLevel	40344	344	Uint	m	0	255	100	R	Measured height of water from the ground.
FEEDBACK	Zone5.AngleLimit	40350	350	Uint	°	0	255	1	R	Maximum angle movement permitted to trackers in automatic mode. This value can be overwritten with a manual setpoint. Value 254 means that DBox in this flood zone is applying its latest received angle limit. Value 255 means that DBox in this flood zone is not applying any angle limit.

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone5.Sensor Code	40351	351	Uint	-	0	3	-	R	Sensor code explaining the quality of the input data. - Value 0 stands for "OK" - Value 1 stands for "No sensor data" - Value 2 stands for "Wrong sensor data" - Value 3 stands for "Waiting for sensor data, applying last angle limit"
FEEDBACK	Zone5.FloodLimitActive	40352	352	Uint	-	0	1	-	R	Flood limit active states if any movement limit is being applied due to flooding sensor readings. Value 0: "Inactive" Value 1: "Active"
FEEDBACK	Zone5.FloodZoneEnabled	40353	353	Uint	-	0	1	-	R	Flood zone enabled states if this flood zone is enabled for limiting angle movement due to flooding sensor readings. Value 0: "Disabled" Value 1: "Enabled"
FEEDBACK	Zone5.FloodLevel	40354	354	Uint	m	0	255	100	R	Measured height of water from the ground.

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone6.AngleLimit	40360	360	Uint	°	0	255	1	R	Maximum angle movement permitted to trackers in automatic mode. This value can be overwritten with a manual setpoint. Value 254 means that DBox in this flood zone is applying its latest received angle limit. Value 255 means that DBox in this flood zone is not applying any angle limit.
FEEDBACK	Zone6.Sensor Code	40361	361	Uint	-	0	3	-	R	Sensor code explaining the quality of the input data. - Value 0 stands for "OK" - Value 1 stands for "No sensor data" - Value 2 stands for "Wrong sensor data" - Value 3 stands for "Waiting for sensor data, applying last angle limit"
FEEDBACK	Zone6.FloodLimitActive	40362	362	Uint	-	0	1	-	R	Flood limit active states if any movement limit is being applied due to flooding sensor readings. Value 0: "Inactive" Value 1: "Active"

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone6.FloodZoneEnabled	40363	363	Uint	-	0	1	-	R	Flood zone enabled states if this flood zone is enabled for limiting angle movement due to flooding sensor readings. Value 0: "Disabled" Value 1: "Enabled"
FEEDBACK	Zone6.FloodLevel	40364	364	Uint	m	0	255	100	R	Measured height of water from the ground.
FEEDBACK	Zone7.AngleLimit	40370	370	Uint	°	0	255	1	R	Maximum angle movement permitted to trackers in automatic mode. This value can be overwritten with a manual setpoint. Value 254 means that DBox in this flood zone is applying its latest received angle limit. Value 255 means that DBox in this flood zone is not applying any angle limit.

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone7.Sensor Code	40371	371	Uint	-	0	3	-	R	Sensor code explaining the quality of the input data. - Value 0 stands for "OK" - Value 1 stands for "No sensor data" - Value 2 stands for "Wrong sensor data" - Value 3 stands for "Waiting for sensor data, applying last angle limit"
FEEDBACK	Zone7.FloodLimitActive	40372	372	Uint	-	0	1	-	R	Flood limit active states if any movement limit is being applied due to flooding sensor readings. Value 0: "Inactive" Value 1: "Active"
FEEDBACK	Zone7.FloodZoneEnabled	40373	373	Uint	-	0	1	-	R	Flood zone enabled states if this flood zone is enabled for limiting angle movement due to flooding sensor readings. Value 0: "Disabled" Value 1: "Enabled"
FEEDBACK	Zone7.FloodLevel	40374	374	Uint	m	0	255	100	R	Measured height of water from the ground.

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone8.AngleLimit	40380	380	Uint	°	0	255	1	R	Maximum angle movement permitted to trackers in automatic mode. This value can be overwritten with a manual setpoint. Value 254 means that DBox in this flood zone is applying its latest received angle limit. Value 255 means that DBox in this flood zone is not applying any angle limit.
FEEDBACK	Zone8.Sensor Code	40381	381	Uint	-	0	3	-	R	Sensor code explaining the quality of the input data. - Value 0 stands for "OK" - Value 1 stands for "No sensor data" - Value 2 stands for "Wrong sensor data" - Value 3 stands for "Waiting for sensor data, applying last angle limit"
FEEDBACK	Zone8.FloodLimitActive	40382	382	Uint	-	0	1	-	R	Flood limit active states if any movement limit is being applied due to flooding sensor readings. Value 0: "Inactive" Value 1: "Active"

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone8.FloodZoneEnabled	40383	383	Uint	-	0	1	-	R	Flood zone enabled states if this flood zone is enabled for limiting angle movement due to flooding sensor readings. Value 0: "Disabled" Value 1: "Enabled"
FEEDBACK	Zone8.FloodLevel	40384	384	Uint	m	0	255	100	R	Measured height of water from the ground.
FEEDBACK	Zone9.AngleLimit	40390	390	Uint	°	0	255	1	R	Maximum angle movement permitted to trackers in automatic mode. This value can be overwritten with a manual setpoint. Value 254 means that DBox in this flood zone is applying its latest received angle limit. Value 255 means that DBox in this flood zone is not applying any angle limit.

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone9.Sensor Code	40391	391	Uint	-	0	3	-	R	Sensor code explaining the quality of the input data. - Value 0 stands for "OK" - Value 1 stands for "No sensor data" - Value 2 stands for "Wrong sensor data" - Value 3 stands for "Waiting for sensor data, applying last angle limit"
FEEDBACK	Zone9.FloodLimitActive	40392	392	Uint	-	0	1	-	R	Flood limit active states if any movement limit is being applied due to flooding sensor readings. Value 0: "Inactive" Value 1: "Active"
FEEDBACK	Zone9.FloodZoneEnabled	40393	393	Uint	-	0	1	-	R	Flood zone enabled states if this flood zone is enabled for limiting angle movement due to flooding sensor readings. Value 0: "Disabled" Value 1: "Enabled"
FEEDBACK	Zone9.FloodLevel	40394	394	Uint	m	0	255	100	R	Measured height of water from the ground.

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone10.AngleLimit	40400	400	Uint	°	0	255	1	R	Maximum angle movement permitted to trackers in automatic mode. This value can be overwritten with a manual setpoint. Value 254 means that DBox in this flood zone is applying its latest received angle limit. Value 255 means that DBox in this flood zone is not applying any angle limit.
FEEDBACK	Zone10.Sensor Code	40401	401	Uint	-	0	3	-	R	Sensor code explaining the quality of the input data. - Value 0 stands for "OK" - Value 1 stands for "No sensor data" - Value 2 stands for "Wrong sensor data" - Value 3 stands for "Waiting for sensor data, applying last angle limit"
FEEDBACK	Zone10.FloodLimitActive	40402	402	Uint	-	0	1	-	R	Flood limit active states if any movement limit is being applied due to flooding sensor readings. Value 0: "Inactive" Value 1: "Active"

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone10.FloodZoneEnabled	40403	403	Uint	-	0	1	-	R	Flood zone enabled states if this flood zone is enabled for limiting angle movement due to flooding sensor readings. Value 0: "Disabled" Value 1: "Enabled"
FEEDBACK	Zone10.FloodLevel	40404	404	Uint	m	0	255	100	R	Measured height of water from the ground.
FEEDBACK	Zone11.AngleLimit	40410	410	Uint	°	0	255	1	R	Maximum angle movement permitted to trackers in automatic mode. This value can be overwritten with a manual setpoint. Value 254 means that DBox in this flood zone is applying its latest received angle limit. Value 255 means that DBox in this flood zone is not applying any angle limit.

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone11.Sensor Code	40411	411	Uint	-	0	3	-	R	Sensor code explaining the quality of the input data. - Value 0 stands for "OK" - Value 1 stands for "No sensor data" - Value 2 stands for "Wrong sensor data" - Value 3 stands for "Waiting for sensor data, applying last angle limit"
FEEDBACK	Zone11.FloodLimitActive	40412	412	Uint	-	0	1	-	R	Flood limit active states if any movement limit is being applied due to flooding sensor readings. Value 0: "Inactive" Value 1: "Active"
FEEDBACK	Zone11.FloodZoneEnabled	40413	413	Uint	-	0	1	-	R	Flood zone enabled states if this flood zone is enabled for limiting angle movement due to flooding sensor readings. Value 0: "Disabled" Value 1: "Enabled"
FEEDBACK	Zone11.FloodLevel	40414	414	Uint	m	0	255	100	R	Measured height of water from the ground.

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone12.AngleLimit	40420	420	Uint	°	0	255	1	R	Maximum angle movement permitted to trackers in automatic mode. This value can be overwritten with a manual setpoint. Value 254 means that DBox in this flood zone is applying its latest received angle limit. Value 255 means that DBox in this flood zone is not applying any angle limit.
FEEDBACK	Zone12.Sensor Code	40421	421	Uint	-	0	3	-	R	Sensor code explaining the quality of the input data. - Value 0 stands for "OK" - Value 1 stands for "No sensor data" - Value 2 stands for "Wrong sensor data" - Value 3 stands for "Waiting for sensor data, applying last angle limit"
FEEDBACK	Zone12.FloodLimitActive	40422	422	Uint	-	0	1	-	R	Flood limit active states if any movement limit is being applied due to flooding sensor readings. Value 0: "Inactive" Value 1: "Active"

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone12.FloodZoneEnabled	40423	423	Uint	-	0	1	-	R	Flood zone enabled states if this flood zone is enabled for limiting angle movement due to flooding sensor readings. Value 0: "Disabled" Value 1: "Enabled"
FEEDBACK	Zone12.FloodLevel	40424	424	Uint	m	0	255	100	R	Measured height of water from the ground.
FEEDBACK	Zone13.AngleLimit	40430	430	Uint	°	0	255	1	R	Maximum angle movement permitted to trackers in automatic mode. This value can be overwritten with a manual setpoint. Value 254 means that DBox in this flood zone is applying its latest received angle limit. Value 255 means that DBox in this flood zone is not applying any angle limit.

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone13.Sensor Code	40431	431	Uint	-	0	3	-	R	Sensor code explaining the quality of the input data. - Value 0 stands for "OK" - Value 1 stands for "No sensor data" - Value 2 stands for "Wrong sensor data" - Value 3 stands for "Waiting for sensor data, applying last angle limit"
FEEDBACK	Zone13.FloodLimitActive	40432	432	Uint	-	0	1	-	R	Flood limit active states if any movement limit is being applied due to flooding sensor readings. Value 0: "Inactive" Value 1: "Active"
FEEDBACK	Zone13.FloodZoneEnabled	40433	433	Uint	-	0	1	-	R	Flood zone enabled states if this flood zone is enabled for limiting angle movement due to flooding sensor readings. Value 0: "Disabled" Value 1: "Enabled"
FEEDBACK	Zone13.FloodLevel	40434	434	Uint	m	0	255	100	R	Measured height of water from the ground.

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone14.AngleLimit	40440	440	Uint	°	0	255	1	R	Maximum angle movement permitted to trackers in automatic mode. This value can be overwritten with a manual setpoint. Value 254 means that DBox in this flood zone is applying its latest received angle limit. Value 255 means that DBox in this flood zone is not applying any angle limit.
FEEDBACK	Zone14.Sensor Code	40441	441	Uint	-	0	3	-	R	Sensor code explaining the quality of the input data. - Value 0 stands for "OK" - Value 1 stands for "No sensor data" - Value 2 stands for "Wrong sensor data" - Value 3 stands for "Waiting for sensor data, applying last angle limit"
FEEDBACK	Zone14.FloodLimitActive	40442	442	Uint	-	0	1	-	R	Flood limit active states if any movement limit is being applied due to flooding sensor readings. Value 0: "Inactive" Value 1: "Active"

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone14.FloodZoneEnabled	40443	443	Uint	-	0	1	-	R	Flood zone enabled states if this flood zone is enabled for limiting angle movement due to flooding sensor readings. Value 0: "Disabled" Value 1: "Enabled"
FEEDBACK	Zone14.FloodLevel	40444	444	Uint	m	0	255	100	R	Measured height of water from the ground.
FEEDBACK	Zone15.AngleLimit	40450	450	Uint	°	0	255	1	R	Maximum angle movement permitted to trackers in automatic mode. This value can be overwritten with a manual setpoint. Value 254 means that DBox in this flood zone is applying its latest received angle limit. Value 255 means that DBox in this flood zone is not applying any angle limit.

FLOOD										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNIT	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Zone15.Sensor Code	40451	451	Uint	-	0	3	-	R	Sensor code explaining the quality of the input data. - Value 0 stands for "OK" - Value 1 stands for "No sensor data" - Value 2 stands for "Wrong sensor data" - Value 3 stands for "Waiting for sensor data, applying last angle limit"
FEEDBACK	Zone15.FloodLimitActive	40452	452	Uint	-	0	1	-	R	Flood limit active states if any movement limit is being applied due to flooding sensor readings. Value 0: "Inactive" Value 1: "Active"
FEEDBACK	Zone15.FloodZoneEnabled	40453	453	Uint	-	0	1	-	R	Flood zone enabled states if this flood zone is enabled for limiting angle movement due to flooding sensor readings. Value 0: "Disabled" Value 1: "Enabled"
FEEDBACK	Zone15.FloodLevel	40454	454	Uint	m	0	255	100	R	Measured height of water from the ground.

NOTICE



Wind data

Wind data for TBox5 and subsequent TBox versions can be found at the corresponding Sensor signal converter section in ID 1 to 254 - Power Block Map (► 50). The sensors installed in the same tower as TBox are located in the same block id as TBox.

6.2.3. ID 1 to 254 - Power Block Map

GENERAL BLOCK CONTROL										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
CONTROL	GlobalMessageCounter	40000	0	Uint	-	0	65535	-	RW	Auto incremented counter when a message is processed by TBox. It does not mean that the message is sent to or received by DBox since LoRa communications are not immediate. Status or setpoint feedback will be available once DBox sends its state.
CONTROL	WindAlarmTrigger	40001	1	Uint	-	0	1	-	RW	Flag signal to send all trackers a global wind alarm.
CONTROL	StopAlarmTrigger	40002	2	Uint	-	0	1	-	RW	Flag signal to send all trackers a global stop alarm.
CONTROL	ManualElevationTargetTrigger	40003	3	Uint	-	0	1	-	RW	Flag signal to send all trackers a global manual setpoint.

GENERAL BLOCK CONTROL										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
CONTROL	ManualElevationTarget	40004	4	Int	°	-7000	7000	100	RW	Global manual setpoint to be sent.
CONTROL	SetAutoControlTrigger	40005	5	Uint	-	0	1	-	RW	Flag signal to send all trackers a set to auto signal.
CONTROL	ConstructionTrigger	40006	6	Uint	-	0	1	-	RW	Flag signal to send all trackers a set to construction position signal.

METEO 0										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	WindSpeed3s	40020	20	Uint	m/s	0	4500	100	R	Average numeric value of wind speed of the last 3 seconds
FEEDBACK	WindDirection3s	40021	21	Uint	°	0	360	1	R	Average numeric value of wind direction of the last 3 seconds
FEEDBACK	WindSpeed30s	40022	22	Uint	m/s	0	4500	100	R	Average numeric value of wind speed of the last 30 seconds
FEEDBACK	WindDirection30s	40023	23	Uint	°	0	360	1	R	Average numeric value of wind direction of the last 30 seconds
FEEDBACK	BatteryLevel	40024	24	Uint	%	0	10000	100	R	Battery level
FEEDBACK	Alarms	40025	25	Uint	-	0	65535	-	R	Alarms: - Bit 0: Anemometer Fault - Bit 1: Charger Fault

METEO 0										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Ew3s	40026	26	Uint	-	1	2	-	RW	Numeric value average of wind direction east/west each 3 seconds 1 - East 2 - West
FEEDBACK	Ew30s	40027	27	Uint	-	1	2	-	RW	Numeric value average of wind direction east/west each 30 seconds 1 - East 2 - West
FEEDBACK	Notifications	40028	28	Uint	-	0	65 535	-	R	Notifications: Bit 0: Outdated
FEEDBACK	WindSpeedEffective	40029	29	Uint	m/s	0	4500	100	R	Effective wind speed. The wind speed value calculated by reducing the weight of the North-South wind component relative to the total wind speed

DBox 1										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
CONTROL	ControlTriggersWord	40050	50	Uint	-	0	65 535	-	RW	
CONTROL	ManualElevationTarget	40051	51	Int	°	-7000	7000	100	RW	
FEEDBACK	OperationMode	40052	52	Uint	-	0	65 535	-	R	
FEEDBACK	ElevationSetpoint	40053	53	Int	°	-7000	7000	100	R	
FEEDBACK	ElevationPosition	40054	54	Int	°	-7000	7000	100	R	
FEEDBACK	BatteryLevel	40055	55	Uint	%	0	10 000	100	R	Refer to Tracker Registers Description: section Single Tracker Register Description [► 83]
FEEDBACK	Alarms	40056	56	Uint	-	0	65 535	-	R	
FEEDBACK	MaxMotorCurrent	40057	57	Uint	mA	0	20 000	1	R	
FEEDBACK	Warnings	40058	58	Uint	-	0	65 535	-	R	
FEEDBACK	PositionCode	40059	59	Uint	-	0	65 535	-	R	
FEEDBACK	Notifications	40060	60	Uint	-	0	65 535	-	R	

DBox 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
CONTROL	ControlTriggersWord	40066	66	Uint	-	0	65 535	-	RW	
CONTROL	ManualElevationTarget	40067	67	Int	°	-7000	7000	100	RW	
FEEDBACK	OperationMode	40068	68	Uint	-	0	65 535	-	R	
FEEDBACK	ElevationSetpoint	40069	69	Int	°	-7000	7000	100	R	
FEEDBACK	ElevationPosition	40070	70	Int	°	-7000	7000	100	R	
FEEDBACK	BatteryLevel	40071	71	Uint	%	0	10 000	100	R	Refer to Tracker Registers Description: section Single Tracker Register Description [► 83]
FEEDBACK	Alarms	40072	72	Uint	-	0	65 535	-	R	
FEEDBACK	MaxMotorCurrent	40073	73	Uint	mA	0	20 000	1	R	
FEEDBACK	Warnings	40074	74	Uint	-	0	65 535	-	R	
FEEDBACK	PositionCode	40075	75	Uint	-	0	65 535	-	R	
FEEDBACK	Notifications	40076	76	Uint	-	0	65 535	-	R	

DBox 3										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
CONTROL	ControlTriggersWord	40082	82	Uint	-	0	65535	-	RW	Refer to Tracker Registers Description: section Single Tracker Register Description [► 83]
CONTROL	ManualElevationTarget	40083	83	Int	°	-7000	7000	100	RW	
FEEDBACK	OperationMode	40084	84	Uint	-	0	65535	-	R	
FEEDBACK	ElevationSetpoint	40085	85	Int	°	-7000	7000	100	R	
FEEDBACK	ElevationPosition	40086	86	Int	°	-7000	7000	100	R	
FEEDBACK	BatteryLevel	40087	87	Uint	%	0	10000	100	R	
FEEDBACK	Alarms	40088	88	Uint	-	0	65535	-	R	
FEEDBACK	MaxMotorCurrent	40089	89	Uint	mA	0	20000	1	R	
FEEDBACK	Warnings	40090	90	Uint	-	0	65535	-	R	
FEEDBACK	PositionCode	40091	91	Uint	-	0	65535	-	R	
FEEDBACK	Notifications	40092	92	Uint	-	0	65535	-	R	

DBox 4										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
CONTROL	ControlTriggersWord	40098	98	Uint	-	0	65535	-	RW	Refer to Tracker Registers Description: section Single Tracker Register Description [► 83]
CONTROL	ManualElevationTarget	40099	99	Int	°	-7000	7000	100	RW	
FEEDBACK	OperationMode	40100	100	Uint	-	0	65535	-	R	

DBox 4										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	ElevationSetpoint	40101	101	Int	°	-7000	7000	100	R	Refer to Tracker Registers Description: section Single Tracker Register Description (► 83)
FEEDBACK	ElevationPosition	40102	102	Int	°	-7000	7000	100	R	
FEEDBACK	BatteryLevel	40103	103	Uint	%	0	10000	100	R	
FEEDBACK	Alarms	40104	104	Uint	-	0	65535	-	R	
FEEDBACK	MaxMotorCurrent	40105	105	Uint	mA	0	20000	1	R	
FEEDBACK	Warnings	40106	106	Uint	-	0	65535	-	R	
FEEDBACK	PositionCode	40107	107	Uint	-	0	65535	-	R	
FEEDBACK	Notifications	40108	108	Uint	-	0	65535	-	R	

DBox 5-500										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
CONTROL	ControlTriggersWord	...	...	Uint	-	0	65535	-	RW	Up to 500 with each of their registers. Refer to Tracker Registers Description: section Single Tracker Register Description (► 83)
CONTROL	ManualElevationTarget	...	...	Int	°	-7000	7000	100	RW	
FEEDBACK	OperationMode	...	...	Uint	-	0	65535	-	R	
FEEDBACK	ElevationSetpoint	...	...	Int	°	-7000	7000	100	R	
FEEDBACK	ElevationPosition	...	...	Int	°	-7000	7000	100	R	

DBox 5-500										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	BatteryLevel	...	...	Uint	%	0	10000	100	R	Up to 500 with each of their registers. Refer to Tracker Registers Description: section Single Tracker Register Description [► 83]
FEEDBACK	Alarms	...	...	Uint	-	0	65535	-	R	
FEEDBACK	MaxMotorCurrent	...	...	Uint	mA	0	20000	1	R	
FEEDBACK	Warnings	...	...	Uint	-	0	65535	-	R	
FEEDBACK	PositionCode	...	...	Uint	-	0	65535	-	R	
FEEDBACK	Notifications	...	...	Uint	-	0	65535	-	R	

GATEWAY DATA 1										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Status	48050	8050	Uint	-	0	65535	-	R	Feedback from Gateway state 1 – No communication 2 – No credentials 3 – Login error 4 - OK

GATEWAY DATA 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Status	48075	8075	Uint	-	0	65 535	-	R	Feedback from Gateway state 1 – No communication 2 – No credentials 3 – Login error 4 – OK

GATEWAY DATA 3										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Status	48100	8100	Uint	-	0	65 535	-	R	Feedback from Gateway state 1 – No communication 2 – No credentials 3 – Login error 4 – OK

GATEWAY DATA 4										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Status	48125	8125	Uint	-	0	65 535	-	R	Feedback from Gateway state 1 – No communication 2 – No credentials 3 – Login error 4 – OK

GATEWAY DATA 5										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Status	48150	8150	Uint	-	0	65 535	-	R	Feedback from Gateway state 1 – No communication 2 – No credentials 3 – Login error 4 – OK

TRACKNET 1										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	PanelVoltage	48175	8175	Uint	V	0	65 535	100	R	Voltage of panel
FEEDBACK	OpenDoor	48177	8177	Uint	-	0	1	-	R	Status of door

TRACKNET 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	PanelVoltage	48195	8195	Uint	V	0	65 535	100	R	Voltage of panel
FEEDBACK	OpenDoor	48197	8197	Uint	-	0	1	-	R	Status of door

TRACKNET 3										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	PanelVoltage	48215	8215	Uint	V	0	65 535	100	R	Voltage of panel
FEEDBACK	OpenDoor	48217	8217	Uint	-	0	1	-	R	Status of door

TRACKNET 4										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	PanelVoltage	48235	8235	Uint	V	0	65 535	100	R	Voltage of panel
FEEDBACK	OpenDoor	48237	8237	Uint	-	0	1	-	R	Status of door

TRACKNET 5										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	PanelVoltage	48255	8255	Uint	V	0	65 535	100	R	Voltage of panel
FEEDBACK	OpenDoor	48257	8257	Uint	-	0	1	-	R	Status of door

METEO 1										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	WindSpeed3s	48275	8275	Uint	m/s	0	4500	100	R	Average numeric value of wind speed of the last 3 seconds
FEEDBACK	WindDirection3s	48276	8276	Uint	°	0	360	1	R	Average numeric value of wind direction of the last 3 seconds
FEEDBACK	WindSpeed30s	48277	8277	Uint	m/s	0	4500	100	R	Average numeric value of wind speed of the last 30 seconds
FEEDBACK	WindDirection30s	48278	8278	Uint	°	0	360	1	R	Average numeric value of wind direction of the last 30 seconds
FEEDBACK	BatteryLevel	48279	8279	Uint	%	0	10 000	100	R	Battery level
FEEDBACK	Alarms	48280	8280	Uint	-	0	65 535	-	R	Alarms: - Bit 0: Anemometer Fault - Bit 1: Charger Fault
FEEDBACK	Ew3s	48281	8281	Uint	-	1	2	-	RW	Numeric value average of wind direction east/west each 3 seconds 1 - East 2 - West
FEEDBACK	Ew30s	48282	8282	Uint	-	1	2	-	RW	Numeric value average of wind direction east/west each 30 seconds 1 - East 2 - West
FEEDBACK	Notifications	48283	8283	Uint	-	0	65 535	-	R	Notifications: Bit 0: Outdated

METEO 1										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	WindSpeedEffective	48284	8284	Uint	m/s	0	4500	100	R	Effective wind speed. The wind speed value calculated by reducing the weight of the North-South wind component relative to the total wind speed

METEO 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	WindSpeed3s	48305	8305	Uint	m/s	0	4500	100	R	Average numeric value of wind speed of the last 3 seconds
FEEDBACK	WindDirection3s	48306	8306	Uint	°	0	360	1	R	Average numeric value of wind direction of the last 3 seconds
FEEDBACK	WindSpeed30s	48307	8307	Uint	m/s	0	4500	100	R	Average numeric value of wind speed of the last 30 seconds
FEEDBACK	WindDirection30s	48308	8308	Uint	°	0	360	1	R	Average numeric value of wind direction of the last 30 seconds
FEEDBACK	BatteryLevel	48309	8309	Uint	%	0	10000	100	R	Battery level
FEEDBACK	Alarms	48310	8310	Uint	-	0	65535	-	R	Alarms: - Bit 0: Anemometer Fault - Bit 1: Charger Fault

METEO 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Ew3s	48311	8311	Uint	-	1	2	-	RW	Numeric value average of wind direction east/west each 3 seconds 1 - East 2 - West
FEEDBACK	Ew30s	48312	8312	Uint	-	1	2	-	RW	Numeric value average of wind direction east/west each 30 seconds 1 - East 2 - West
FEEDBACK	Notifications	48313	8313	Uint	-	0	65 535	-	R	Notifications: Bit 0: Outdated
FEEDBACK	WindSpeedEffective	48314	8314	Uint	m/s	0	4500	100	R	Effective wind speed. The wind speed value calculated by reducing the weight of the North-South wind component relative to the total wind speed

METEO 3										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	WindSpeed3s	48335	8335	Uint	m/s	0	4500	100	R	Average numeric value of wind speed of the last 3 seconds

METEO 3										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	WindDirection3s	48336	8336	Uint	°	0	360	1	R	Average numeric value of wind direction of the last 3 seconds
FEEDBACK	WindSpeed30s	48337	8337	Uint	m/s	0	4500	100	R	Average numeric value of wind speed of the last 30 seconds
FEEDBACK	WindDirection30s	48338	8338	Uint	°	0	360	1	R	Average numeric value of wind direction of the last 30 seconds
FEEDBACK	BatteryLevel	48339	8339	Uint	%	0	10 000	100	R	Battery level
FEEDBACK	Alarms	48340	8340	Uint	-	0	65 535	-	R	Alarms: - Bit 0: Anemometer Fault - Bit 1: Charger Fault
FEEDBACK	Ew3s	48341	8341	Uint	-	1	2	-	RW	Numeric value average of wind direction east/west each 3 seconds 1 - East 2 - West
FEEDBACK	Ew30s	48342	8342	Uint	-	1	2	-	RW	Numeric value average of wind direction east/west each 30 seconds 1 - East 2 - West
FEEDBACK	Notifications	48343	8343	Uint	-	0	65 535	-	R	Notifications: Bit 0: Outdated

METEO 3										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	WindSpeedEffective	48344	8344	Uint	m/s	0	4500	100	R	Effective wind speed. The wind speed value calculated by reducing the weight of the North-South wind component relative to the total wind speed

METEO 4										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	WindSpeed3s	48365	8365	Uint	m/s	0	4500	100	R	Average numeric value of wind speed of the last 3 seconds
FEEDBACK	WindDirection3s	48366	8366	Uint	°	0	360	1	R	Average numeric value of wind direction of the last 3 seconds
FEEDBACK	WindSpeed30s	48367	8367	Uint	m/s	0	4500	100	R	Average numeric value of wind speed of the last 30 seconds
FEEDBACK	WindDirection30s	48368	8368	Uint	°	0	360	1	R	Average numeric value of wind direction of the last 30 seconds
FEEDBACK	BatteryLevel	48369	8369	Uint	%	0	10000	100	R	Battery level
FEEDBACK	Alarms	48370	8370	Uint	-	0	65535	-	R	Alarms: - Bit 0: Anemometer Fault - Bit 1: Charger Fault

METEO 4										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Ew3s	48371	8371	Uint	-	1	2	-	RW	Numeric value average of wind direction east/west each 3 seconds 1 - East 2 - West
FEEDBACK	Ew30s	48372	8372	Uint	-	1	2	-	RW	Numeric value average of wind direction east/west each 30 seconds 1 - East 2 - West
FEEDBACK	Notifications	48373	8373	Uint	-	0	65 535	-	R	Notifications: Bit 0: Outdated
FEEDBACK	WindSpeedEffective	48374	8374	Uint	m/s	0	4500	100	R	Effective wind speed. The wind speed value calculated by reducing the weight of the North-South wind component relative to the total wind speed

METEO 5										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	WindSpeed3s	48395	8395	Uint	m/s	0	4500	100	R	Average numeric value of wind speed of the last 3 seconds
FEEDBACK	WindDirection3s	48396	8396	Uint	°	0	360	1	R	Average numeric value of wind direction of the last 3 seconds
FEEDBACK	WindSpeed30s	48397	8397	Uint	m/s	0	4500	100	R	Average numeric value of wind speed of the last 30 seconds
FEEDBACK	WindDirection30s	48398	8398	Uint	°	0	360	1	R	Average numeric value of wind direction of the last 30 seconds
FEEDBACK	BatteryLevel	48399	8399	Uint	%	0	10000	100	R	Battery level
FEEDBACK	Alarms	48400	8400	Uint	-	0	65535	-	R	Alarms: - Bit 0: Anemometer Fault - Bit 1: Charger Fault
FEEDBACK	Ew3s	48401	8401	Uint	-	1	2	-	RW	Numeric value average of wind direction east/west each 3 seconds 1 - East 2 - West
FEEDBACK	Ew30s	48402	8402	Uint	-	1	2	-	RW	Numeric value average of wind direction east/west each 30 seconds 1 - East 2 - West
FEEDBACK	Notifications	48403	8403	Uint	-	0	65535	-	R	Notifications: Bit 0: Outdated

METEO 5										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	WindSpeedEffective	48404	8404	Uint	m/s	0	4500	100	R	Effective wind speed. The wind speed value calculated by reducing the weight of the North-South wind component relative to the total wind speed

SENSOR SIGNAL CONVERTER 1 - WIND SENSOR 1										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	SensorError	48425	8425	Variable	-	0	1	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	WindSpeedReal	48426	8426	Variable	m/s	0	4500	100	R	The most recent real-time measured value of wind speed
FEEDBACK	WindDirectionReal	48427	8427	Variable	°	0	360	1	R	The most recent real-time measured value of wind direction
FEEDBACK	WindDirectionEWR	48428	8428	Variable	-	-	-	-	R	Latest real time measurement value of wind direction east/west (1 -East) (2 - West)
FEEDBACK	WindSpeedShortAvg	48429	8429	Variable	m/s	0	4500	100	R	Numeric value average of wind speed every 3 seconds
FEEDBACK	WindDirectionShortAvg	48430	8430	Variable	°	0	360	1	R	Numeric value average of wind direction every 3 seconds

SENSOR SIGNAL CONVERTER 1 - WIND SENSOR 1										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	WindDirectionEWS hortAvg	48431	8431	Variable	-	-	-	-	R	Numeric value average of wind direction east/west every 3 seconds (1 -East) (2 - West)
FEEDBACK	WindSpeedLongAvg	48432	8432	Variable	m/s	0	4500	100	R	Numeric value average of wind speed every 30 seconds
FEEDBACK	WindDirectionLong Avg	48433	8433	Variable	°	0	360	1	R	Numeric value average of wind direction every 30 seconds
FEEDBACK	WindDirectionEWL ongAvg	48434	8434	Variable	-	-	-	-	R	Numeric value average of wind direction east/west every 30 seconds (1 -East) (2 - West)
FEEDBACK	WindSpeedGust	48435	8435	Variable	m/s	0	4500	100	R	Numeric value of the maximum moving average of wind speed for 3 seconds, of the last 10 seconds.
FEEDBACK	WindDirectionGust	48436	8436	Variable	°	0	360	1	R	Numeric value average of wind direction every 3 seconds
FEEDBACK	WindDirectionEWG ust	48437	8437	Variable	-	-	-	-	R	Numeric value average of wind direction east/west every 3 seconds (1 -East) (2 - West)
FEEDBACK	WindSpeedEffective	48438	8438	Uint	m/s	0	4500	100	R	Effective wind speed. The wind speed value calculated by reducing the weight of the North-South wind component relative to the total wind speed

SENSOR SIGNAL CONVERTER 1 - WIND SENSOR 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	SensorError	48455	8455	Variable	-	0	1	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	WindSpeedReal	48456	8456	Variable	m/s	0	4500	100	R	Latest real time measurement value of wind speed
FEEDBACK	WindDirectionReal	48457	8457	Variable	°	0	360	1	R	Latest real time measurement value of wind direction
FEEDBACK	WindDirectionEWR eal	48458	8458	Variable	-	-	-	-	R	Latest real time measurement value of wind direction east/ west (1 -East) (2 - West)
FEEDBACK	WindSpeedShortAv g	48459	8459	Variable	m/s	0	4500	100	R	Numeric value average of wind speed every 3 seconds
FEEDBACK	WindDirectionShort Avg	48460	8460	Variable	°	0	360	1	R	Numeric value average of wind direction every 3 seconds
FEEDBACK	WindDirectionEWS hortAvg	48461	8461	Variable	-	-	-	-	R	Numeric value average of wind direction east/west every 3 seconds (1 - East) (2 - West)
FEEDBACK	WindSpeedLongAv g	48462	8462	Variable	m/s	0	4500	100	R	Numeric value average of wind speed every 30 seconds
FEEDBACK	WindDirectionLong Avg	48463	8463	Variable	°	0	360	1	R	Numeric value average of wind direction every 30 seconds
FEEDBACK	WindDirectionEWL ongAvg	48464	8464	Variable	-	-	-	-	R	Numeric value average of wind direction east/west every 30 seconds (1 - East) (2 - West)

SENSOR SIGNAL CONVERTER 1 - WIND SENSOR 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	WindSpeedGust	48465	8465	Variable	m/s	0	4500	100	R	Numeric value of the maximum moving average of wind speed for 3 seconds, of the last 10 seconds.
FEEDBACK	WindDirectionGust	48466	8466	Variable	°	0	360	1	R	Numeric value average of wind direction every 3 seconds
FEEDBACK	WindDirectionEWGust	48467	8467	Variable	-	-	-	-	R	Numeric value average of wind direction east/west every 3 seconds (1-East) (2 - West)
FEEDBACK	WindSpeedEffective	48468	8468	Uint	m/s	0	4500	100	R	Effective wind speed. The wind speed value calculated by reducing the weight of the North-South wind component relative to the total wind speed

SENSOR SIGNAL CONVERTER 2 - WIND SENSOR 1										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	SensorError	48485	8485	Variable	-	0	1	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	WindSpeedReal	48486	8486	Variable	m/s	0	4500	100	R	Latest real time measurement value of wind speed
FEEDBACK	WindDirectionReal	48487	8487	Variable	°	0	360	1	R	Latest real time measurement value of wind direction

SENSOR SIGNAL CONVERTER 2 - WIND SENSOR 1										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	WindDirectionEWR eal	48488	8488	Vari able	-	-	-	-	R	Latest real time measurement value of wind direction east/ west (1 - East) (2 - West)
FEEDBACK	WindSpeedShortAv g	48489	8489	Vari able	m/s	0	4500	100	R	Numeric value average of wind speed every 3 seconds
FEEDBACK	WindDirectionShort Avg	48490	8490	Vari able	°	0	360	1	R	Numeric value average of wind direction every 3 seconds
FEEDBACK	WindDirectionEWS hortAvg	48491	8491	Vari able	-	-	-	-	R	Numeric value average of wind direction east/west every 3 seconds (1 - East) (2 - West)
FEEDBACK	WindSpeedLongAv g	48492	8492	Vari able	m/s	0	4500	100	R	Numeric value average of wind speed every 30 seconds
FEEDBACK	WindDirectionLong Avg	48493	8493	Vari able	°	0	360	1	R	Numeric value average of wind direction every 30 seconds
FEEDBACK	WindDirectionEWL ongAvg	48494	8494	Vari able	-	-	-	-	R	Numeric value average of wind direction east/west every 30 seconds (1 - East) (2 - West)
FEEDBACK	WindSpeedGust	48495	8495	Vari able	m/s	0	4500	100	R	Numeric value of the maximum moving average of wind speed for 3 seconds, of the last 10 seconds.
FEEDBACK	WindDirectionGust	48496	8496	Vari able	°	0	360	1	R	Numeric value average of wind direction every 3 seconds

SENSOR SIGNAL CONVERTER 2 - WIND SENSOR 1										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	WindDirectionEWGust	48497	8497	Variable	-	-	-	-	R	Numeric value average of wind direction east/west every 3 seconds (1 - East) (2 - West)
FEEDBACK	WindSpeedEffective	48498	8498	Uint	m/s	0	4500	100	R	Effective wind speed. The wind speed value calculated by reducing the weight of the North-South wind component relative to the total wind speed

SENSOR SIGNAL CONVERTER 2 - WIND SENSOR 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	SensorError	48515	8515	Variable	-	0	1	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	WindSpeedReal	48516	8516	Variable	m/s	0	4500	100	R	Latest real time measurement value of wind speed
FEEDBACK	WindDirectionReal	48517	8517	Variable	°	0	360	1	R	Latest real time measurement value of wind direction
FEEDBACK	WindDirectionEWR	48518	8518	Variable	-	-	-	-	R	Latest real time measurement value of wind direction east/west (1 -East) (2 - West)
FEEDBACK	WindSpeedShortAvg	48519	8519	Variable	m/s	0	4500	100	R	Numeric value average of wind speed every 3 seconds

SENSOR SIGNAL CONVERTER 2 - WIND SENSOR 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	WindDirectionShortAvg	48520	8520	Variable	°	0	360	1	R	Numeric value average of wind direction every 3 seconds
FEEDBACK	WindDirectionEWS hortAvg	48521	8521	Variable	-	-	-	-	R	Numeric value average of wind direction east/west every 3 seconds (1 -East) (2 - West)
FEEDBACK	WindSpeedLongAvg	48522	8522	Variable	m/s	0	4500	100	R	Numeric value average of wind speed every 30 seconds
FEEDBACK	WindDirectionLongAvg	48523	8523	Variable	°	0	360	1	R	Numeric value average of wind direction every 30 seconds
FEEDBACK	WindDirectionEWL ongAvg	48524	8524	Variable	-	-	-	-	R	Numeric value average of wind direction east/west every 30 seconds (1 - East) (2 - West)
FEEDBACK	WindSpeedGust	48525	8525	Variable	m/s	0	4500	100	R	Numeric value of the maximum moving average of wind speed for 3 seconds, of the last 10 seconds.
FEEDBACK	WindDirectionGust	48526	8526	Variable	°	0	360	1	R	Numeric value average of wind direction every 3 seconds
FEEDBACK	WindDirectionEWG ust	48527	8527	Variable	-	-	-	-	R	Numeric value average of wind direction east/west every 3 seconds (1 - East) (2 - West)
FEEDBACK	WindSpeedEffective	48528	8528	Uint	m/s	0	4500	100	R	Effective wind speed. The wind speed value calculated by reducing the weight of the North-South wind component relative to the total wind speed

SENSOR SIGNAL CONVERTER 2 - WIND SENSOR 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	SensorError	48545	8545	Variable	-	0	1	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	WindSpeedReal	48546	8546	Variable	m/s	0	4500	100	R	Latest real time measurement value of wind speed
FEEDBACK	WindDirectionReal	48547	8547	Variable	°	0	360	1	R	Latest real time measurement value of wind direction
FEEDBACK	WindDirectionEWR eal	48548	8548	Variable	-	-	-	-	R	Latest real time measurement value of wind direction east/ west (1 - East) (2 - West)
FEEDBACK	WindSpeedShortAv g	48549	8549	Variable	m/s	0	4500	100	R	Numeric value average of wind speed every 3 seconds
FEEDBACK	WindDirectionShort Avg	48550	8550	Variable	°	0	360	1	R	Numeric value average of wind direction every 3 seconds
FEEDBACK	WindDirectionEWS hortAvg	48551	8551	Variable	-	-	-	-	R	Numeric value average of wind direction east/west every 3 seconds (1 -East) (2 - West)
FEEDBACK	WindSpeedLongAv g	48552	8552	Variable	m/s	0	4500	100	R	Numeric value average of wind speed every 30 seconds
FEEDBACK	WindDirectionLong Avg	48553	8553	Variable	°	0	360	1	R	Numeric value average of wind direction every 30 seconds
FEEDBACK	WindDirectionEWL ongAvg	48554	8554	Variable	-	-	-	-	R	Numeric value average of wind direction east/west every 30 seconds (1 - East) (2 - West)

SENSOR SIGNAL CONVERTER 2 - WIND SENSOR 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	WindSpeedGust	48555	8555	Variable	m/s	0	4500	100	R	Numeric value of the maximum moving average of wind speed for 3 seconds, of the last 10 seconds.
FEEDBACK	WindDirectionGust	48556	8556	Variable	°	0	360	1	R	Numeric value average of wind direction every 3 seconds
FEEDBACK	WindDirectionEWGust	48557	8557	Variable	-	-	-	-	R	Numeric value average of wind direction east/west every 3 seconds (1 - East) (2 - West)
FEEDBACK	WindSpeedEffective	48558	8558	Uint	m/s	0	4500	100	R	Effective wind speed. The wind speed value calculated by reducing the weight of the North-South wind component relative to the total wind speed
FEEDBACK	SensorError	48575	8575	Variable	-	0	1	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	WindSpeedReal	48576	8576	Variable	m/s	0	4500	100	R	Latest real time measurement value of wind speed
FEEDBACK	WindDirectionReal	48577	8577	Variable	°	0	360	1	R	Latest real time measurement value of wind direction
FEEDBACK	WindDirectionEWR	48578	8578	Variable	-	-	-	-	R	Latest real time measurement value of wind direction east/west (1 -East) (2 - West)
FEEDBACK	WindSpeedShortAvg	48579	8579	Variable	m/s	0	4500	100	R	Numeric value average of wind speed every 3 seconds

SENSOR SIGNAL CONVERTER 2 - WIND SENSOR 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	WindDirectionShortAvg	48580	8580	Variable	°	0	360	1	R	Numeric value average of wind direction every 3 seconds
FEEDBACK	WindDirectionEWS hortAvg	48581	8581	Variable	-	-	-	-	R	Numeric value average of wind direction east/west every 3 seconds (1 - East) (2 - West)
FEEDBACK	WindSpeedLongAvg	48582	8582	Variable	m/s	0	4500	100	R	Numeric value average of wind speed every 30 seconds
FEEDBACK	WindDirectionLongAvg	48583	8583	Variable	°	0	360	1	R	Numeric value average of wind direction every 30 seconds
FEEDBACK	WindDirectionEWLongAvg	48584	8584	Variable	-	-	-	-	R	Numeric value average of wind direction east/west every 30 seconds (1 - East) (2 - West)
FEEDBACK	WindSpeedGust	48585	8585	Variable	m/s	0	4500	100	R	Numeric value of the maximum moving average of wind speed for 3 seconds, of the last 10 seconds.
FEEDBACK	WindDirectionGust	48586	8586	Variable	°	0	360	1	R	Numeric value average of wind direction every 3 seconds
FEEDBACK	WindDirectionEWGust	48587	8587	Variable	-	-	-	-	R	Numeric value average of wind direction east/west every 3 seconds (1 - East) (2 - West)
FEEDBACK	WindSpeedEffective	48588	8588	Uint	m/s	0	4500	100	R	Effective wind speed. The wind speed value calculated by reducing the weight of the North-South wind component relative to the total wind speed

SENSOR SIGNAL CONVERTER 2 - WIND SENSOR 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	SensorError	48605	8605	Variable	-	0	1	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	Temperature	48606	8606	Variable	°C	-10 000	10 000	100	R	Air temperature
FEEDBACK	Humidity	48607	8607	Variable	%	0	10 000	100	R	Air relative humidity
FEEDBACK	Pressure	48608	8608	Variable	mba r	0	2000	1	R	Atmospheric pressure
FEEDBACK	SensorError	48615	8615	Variable	-	0	1	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	Temperature	48616	8616	Variable	°C	-10 000	10 000	100	R	Air temperature
FEEDBACK	Humidity	48617	8617	Variable	%	0	10 000	100	R	Air relative humidity
FEEDBACK	Pressure	48618	8618	Variable	mba r	0	2000	1	R	Atmospheric pressure
FEEDBACK	SensorError	48625	8625	Variable	-	0	1	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	Temperature	48626	8626	Variable	°C	-10 000	10 000	100	R	Air temperature
FEEDBACK	Humidity	48627	8627	Variable	%	0	10 000	100	R	Air relative humidity
FEEDBACK	Pressure	48628	8628	Variable	mba r	0	2000	1	R	Atmospheric pressure

SENSOR SIGNAL CONVERTER 2 - WIND SENSOR 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	SensorError	48635	8635	Variable	-	0	1	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	Temperature	48636	8636	Variable	°C	-10 000	10 000	100	R	Air temperature
FEEDBACK	Humidity	48637	8637	Variable	%	0	10 000	100	R	Air relative humidity
FEEDBACK	Pressure	48638	8638	Variable	mba r	0	2000	1	R	Atmospheric pressure
FEEDBACK	SensorError	48645	8645	Variable	-	0	1	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	Temperature	48646	8646	Variable	°C	-10 000	10 000	100	R	Air temperature
FEEDBACK	Humidity	48647	8647	Variable	%	0	10 000	100	R	Air relative humidity
FEEDBACK	Pressure	48648	8648	Variable	mba r	0	2000	1	R	Atmospheric pressure
FEEDBACK	SensorError	48655	8655	Variable	-	0	1	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	Temperature	48656	8656	Variable	°C	-10 000	10 000	100	R	Air temperature
FEEDBACK	Humidity	48657	8657	Variable	%	0	10 000	100	R	Air relative humidity
FEEDBACK	Pressure	48658	8658	Variable	mba r	0	2000	1	R	Atmospheric pressure

SENSOR SIGNAL CONVERTER 2 - WIND SENSOR 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	SensorError	48665	8665	Variable	-	-	-	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	MaxHailSize	48666	8666	Variable	mm	0	2000	10	R	Maximum hail diameter measured by the sensor.
FEEDBACK	SensorError	48675	8675	Variable	-	-	-	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	MaxHailSize	48676	8676	Variable	mm	0	2000	10	R	Maximum hail diameter measured by the sensor.
FEEDBACK	SensorError	48685	8685	Variable	-	-	-	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	MaxHailSize	48686	8686	Variable	mm	0	2000	10	R	Maximum hail diameter measured by the sensor.
FEEDBACK	SensorError	48695	8695	Variable	-	-	-	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	MaxHailSize	48696	8696	Variable	mm	0	2000	10	R	Maximum hail diameter measured by the sensor.
FEEDBACK	SensorError	48705	8705	Variable	-	-	-	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	MaxHailSize	48706	8706	Variable	mm	0	2000	10	R	Maximum hail diameter measured by the sensor.
FEEDBACK	SensorError	48715	8715	Variable	-	-	-	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	MaxHailSize	48716	8716	Variable	mm	0	2000	10	R	Maximum hail diameter measured by the sensor.

SENSOR SIGNAL CONVERTER 2 - WIND SENSOR 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	SensorError	48725	8725	Variable	-	0	1	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	Height	48726	8726	Variable	mm	-500 0	5000	1	R	Snow column height.
FEEDBACK	Snowing	48727	8727	Variable	-	0	1	-	R	Value 0: Not snowing Value 1: Snowing
FEEDBACK	SensorError	48735	8735	Variable	-	0	1	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	Height	48736	8736	Variable	mm	-500 0	5000	1	R	Snow column height.
FEEDBACK	Snowing	48737	8737	Variable	-	0	1	-	R	Value 0: Not snowing Value 1: Snowing
FEEDBACK	SensorError	48745	8745	Variable	-	0	1	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	Height	48746	8746	Variable	mm	-500 0	5000	1	R	Snow column height.
FEEDBACK	Snowing	48747	8747	Variable	-	0	1	-	R	Value 0: Not snowing Value 1: Snowing
FEEDBACK	SensorError	48755	8755	Variable	-	0	1	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	Height	48756	8756	Variable	mm	-500 0	5000	1	R	Snow column height.
FEEDBACK	Snowing	48757	8757	Variable	-	0	1	-	R	Value 0: Not snowing Value 1: Snowing

SENSOR SIGNAL CONVERTER 2 - WIND SENSOR 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	SensorError	48765	8765	Variable	-	0	1	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	Height	48766	8766	Variable	mm	-500 0	5000	1	R	Snow column height.
FEEDBACK	Snowing	48767	8767	Variable	-	0	1	-	R	Value 0: Not snowing Value 1: Snowing
FEEDBACK	SensorError	48775	8775	Variable	-	0	1	-	R	Value 0: invalid data Value 1: valid data
FEEDBACK	Height	48776	8776	Variable	mm	-500 0	5000	1	R	Snow column height.
FEEDBACK	Snowing	48777	8777	Variable	-	0	1	-	R	Value 0: Not snowing Value 1: Snowing

NOTICE



Wind data

Wind data for TBox versions prior to TBox5 are located at Addresses 40014 and 40015 (Global Status folder) in ID 255 - General TBox Map and Configuration (► 15).

6.2.4. Single Tracker Register Description

System version 3.2

Table 2. Power Block description

TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
Bit Position:									
ControlTriggers Word	-	-	Uint	-	0	65535	-	RW	• 0 = Stop • 1 = Manual • 2 = Auto • 3 = Construction
ManualElevation Target	-	-	Int	°	-7000	7000	100	RW	Manual setpoint to be sent
Operation Mode:									
OperationMode	-	-	Uint	-	0	65535	-	R	• Value 0. Defect: Not recognized • Value 1. TBox Control: Auto • Value 2. Remote Control: Manual Position • Value 3. Local Control: PVH application • Value 4. Halt: Waiting for acknowledge • Value 5. Remote Control: Manual Wind Position • Value 6. Defect: Secure Battery

TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
OperationMode	-	-	Uint	-	0	65535	-	R	- Value 7. Defect: Fault stop - Value 8. Local Control: Emergency stop - Value 9. Remote Control: Manual Stop - Value 10. Defect: TBox communications lost - Value 11. Local Control: Buttons - Value 12. Remote Control: Manual Construction Position
ElevationSetpoint	-	-	Int	Degrees	-7000	7000	100	R	Target elevation in degrees from horizontal. Auto or manual depending on mode of operation
ElevationPosition	-	-	Int	°	-7000	7000	100	R	Actual elevation position in degrees from horizontal
BatteryLevel	-	-	Uint	%	0	10000	100	R	Battery level
Alarms	-	-	Uint	-	0	65535	-	R	Alarms: - Bit 0. Emergency Stop - Bit 1. Charger Fault - Bit 2. Inclinometer Fault - Bit 3. Wrong direction movement - Bit 4. Overcurrent Fault - Bit 5. Slow motor movement - Bit 6. High power consumption

TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
Alarms	-	-	Uint	-	0	65535	-	R	• Bit 7. Static inclinometer read • Bit 8. Out of boundaries • Bit 9. Low battery • Bit 10. Low temperature • Bit 11. High temperature • Bit 12. PV panel fault • Bit 13. Wrong firmware Fault • Bit 14. Angle Step Fault • Bit 15. Maximum movement time
MaxMotorCurrent	-	-	Uint	mA	0	20000	1	R	Maximum motor current of the last performed movement without considering the distance between its first and last angle.
Warnings	-	-	Uint	-	0	65535	-	R	The SoC mode, or State of Charge mode, indicates whether DBox is: • integrating its consumption to calculate its battery level (0 or 0b0000 0000 0000 0000) • using the value retrieved from its power management module (64 or 0b0000 0000 0100 0000) • no battery is found in DBox (0 or 0b0000 0000 1000 0000)

TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
PositionCode	-	-	Uint	-	0	65535	-	R	Position Code retrieves information regarding why DBox is in a position during automatic and nominal operation: • Value 0. Tracking • Value 1. Night position • Value 2. Backtracking • Value 3. Diffuse • Value 4. Reserved • Value 5. Wind defense • Value 6. Construction defense • Value 7. Snow position • Value 8. Hail position • Value 9: User Position • Value 10: Stop position
Notification	-	-	Uint	-	0	65535	-	R	Notification: • Bit 0. Outdated DBox Data • Bit 1. Reserved • Bit 2. Reserved • Bit 3. Reserved • Bit 4. Reserved • Bit 5. Reserved • Bit 6. Reserved

TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
Notification	-	-	Uint	-	0	65535	-	R	• Bit 7. Reserved • Bit 8. Reserved • Bit 9. Reserved • Bit 10. Reserved • Bit 11. Reserved • Bit 12. Reserved • Bit 13. Reserved • Bit 14. Reserved • Bit 15. Reserved

System version 3.1

Table 3. Power Block description

TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
Bit Position:									
ControlTriggers Word	-	-	Uint	-	0	65535	-	RW	• 0 = Stop • 1 = Manual • 2 = Auto • 3 = Construction
ManualElevation Target	-	-	Int	°	-7000	7000	100	RW	Manual setpoint to be sent
Operation Mode:									
OperationMode	-	-	Uint	-	0	65535	-	R	• Value 0. Defect: Not recognized • Value 1. TBox Control: Auto • Value 2. Remote Control: Manual Position • Value 3. Local Control: PVH application • Value 4. Halt: Waiting for acknowledge • Value 5. Remote Control: Manual Wind Position • Value 6. Defect: Secure Battery • Value 7. Defect: Fault stop • Value 8. Local Control: Emergency stop

TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
OperationMode	-	-	Uint	-	0	65535	-	R	- Value 9. Remote Control: Manual Stop - Value 10. Defect: TBox communications lost - Value 11. Local Control: Buttons - Value 12. Remote Control: Manual Construction Position
ElevationSetpoint	-	-	Int	°	-7000	7000	100	R	Target elevation in degrees from horizontal. Auto or manual depending on mode of operation
ElevationPosition	-	-	Int	°	-7000	7000	100	R	Actual elevation position in degrees from horizontal
BatteryLevel	-	-	Uint	%	0	10000	100	R	Battery level
Alarms	-	-	Uint	-	0	65535	-	R	Alarms: - Bit 0. Emergency Stop Button Pressed - Bit 1. Charger Fault - Bit 2. Inclinator Fault - Bit 3. Motor Wrong Direction Fault - Bit 4. Motor Overcurrent - Bit 5. Slow motor - Bit 6. High power consumption - Bit 7. Static Inclinator Read - Bit 8. Angle out of boundaries

TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
Alarms	-	-	Uint	-	0	65535	-	R	• Bit 9. Low battery • Bit 10. Low temperature • Bit 11. High temperature • Bit 12. PV panel voltage fault • Bit 13. Wrong FW Fault • Bit 14. Angle Step Fault • Bit 15. Outdated DBox Data
MaxMotorCurrent	-	-	Uint	mA	0	20000	1	R	Maximum motor current of the last performed movement without considering the distance between its first and last angle.
Warnings	-	-	-	-	-	-	-	-	Not applicable in version 3.1
PositionCode	-	-	-	-	-	-	-	-	Not applicable in version 3.1
Notification	-	-	-	-	-	-	-	-	Not applicable in version 3.1

6.3. Integration examples

6.3.1. ID 255 - General TBox Map and Configuration

Global status

To monitor all the plant remotely, SCADA must be connected to the TBox IP through the 255 Modbus ID to be able to read the variables of the Modbus map. To see Modbus remote global read addresses, please refer to the section Modbus memory map registers (► 13).

Read example: Backtracking Status

- Modbus port: 502
- Device Address: 255
- Function Code: 03
- Register Address: 40011

Global control

To control all the plant remotely, SCADA must be connected to the TBox IP through the 255 Modbus ID to be able to write the command in the addresses of the Modbus map. To see Modbus remote control global addresses, refer to section Modbus memory map registers (► 13).

Write example: Send Wind Alarm to all DBox.

- Modbus port: 502
- Device Address: 255
- Function Code: 06
- Register Address: 40101
- 40101 --> 1

Automode

To configure the algorithms to be executed in the TBox, SCADA must be connected to the TBox IP through the 255 Modbus ID to be able to read and write the command in the addresses of the Modbus Map. To check Modbus automode folder, refer to the section Modbus memory map registers (► 13).

Write example: Enable snow management.

- Modbus port: 502
- Device Address: 255
- Function Code: 06
- Register Address: 40151
- 40151 --> 1

Read example: read the status of the diffuse configuration, is it enabled?

- Modbus port: 502
- Device Address: 255
- Function Code: 03
- Register Address: 40156
- Number of registers: 1

Flood

To monitor Flood zone 5, SCADA must be connected to the TBox IP through the 255 Modbus ID to be able to read the variables of the Modbus map. To see Modbus remote global read addresses, please refer to the section Modbus memory map registers (► 13).

Read example: Backtracking Status

- Modbus port: 502
- Device Address: 255
- Function Code: 03
- Register Address: 40350
- Number of registers: 5

6.3.2. ID 1 to 254 - Power-Block Map

General Block Control

To control all the plant remotely, SCADA must be connected to the TBox IP through the ID of the desired Block to be able to write the command in the addresses of the Modbus map. To see Modbus general block control addresses, refer to section Modbus memory map registers (► 13).

Write example: Send Stop Alarm only to block 7

- Modbus port: 502
- Device Address: 7
- Function Code: 06
- Register Address: 40002
- 40002 --> 1

Meteo 0

To monitor legacy Meteo data in Block 14, SCADA must be connected to the TBox IP through the ID 14. To see Modbus read addresses, refer to section Modbus memory map registers (► 13).

Read example

- Modbus port: 502
- Device Address: 14
- Function Code: 03
- Register Address: 40020
- Number of registers: 8

DBox 1-500

To control a single DBox remotely, SCADA must be connected to the TBox IP through the ID of the desired Block to be able to write the command in the addresses of the Modbus map. To see Modbus DBox Block X control addresses, refer to section Modbus memory map registers (► 13).

Write example: Send Manual command to +30 degrees to a single DBox, block 2 DBox 4

- Modbus port: 502
- Device Address: 2
- Function Code: 16
- Register Address: 40099
- 40099 --> 3000
- Register Address: 40098
- 40098 --> 2 (in order to activate the bit position 1 (DEC=2 BIN=0010))

Read example: Read the current operation mode of a single DBox, block 2 DBox 4

- Modbus port: 502
- Device Address: 5
- Function Code: 03
- Register Address: 40100
- Number of registers: 1

Write example: Send Stop to a single DBox, block 2 DBox 5

- Modbus port: 502
- Device Address: 2
- Function Code: 16
- Register Address: 40114
- 40114 --> 0

Read example: Read all the information of a single DBox, block 5 DBox 3

- Modbus port: 502
- Device Address: 5
- Function Code: 03
- Register Address: 40084
- Number of registers: 6

LoRa Gateway 1-5

To monitor LoRa Gateway 1 in Block 3, SCADA must be connected to the TBox IP through the ID 3. To see Modbus read addresses, refer to section Modbus memory map registers (► 13).

Read example

- Modbus port: 502
- Device Address: 3
- Function Code: 03
- Register Address: 48050
- Number of registers: 1

Tracknet 1-5

To monitor Tracknet 2 in Block 56, SCADA must be connected to the TBox IP through the ID 56. To see Modbus read addresses, refer to section Modbus memory map registers (► 13).

Read example

- Modbus port: 502
- Device Address: 56
- Function Code: 03
- Register Address: 48215
- Number of registers: 3

Meteo 1-5

To monitor Meteo data of element 3 in Block 9, SCADA must be connected to the TBox IP through the ID 9. To see Modbus read addresses, refer to section Modbus memory map registers (► 13).

Read example

- Modbus port: 502
- Device Address: 9
- Function Code: 03
- Register Address: 48335
- Number of registers: 9

Sensor Signal Converters 1-3 Wind Sensors 1-2

To monitor Sensor Signal Converters 3 Wind Sensor 1 in Block 3, SCADA must be connected to the TBox IP through the ID 3. To see Modbus read addresses, refer to section Modbus memory map registers (► 13).

Read example

- Modbus port: 502
- Device Address: 6
- Function Code: 03
- Register Address: 48695
- Number of registers: 2

Sensor Signal Converters 1-3 Air Sensors 1-2

To monitor Sensor Signal Converters 1 Hail Sensor 2 in Block 6, SCADA must be connected to the TBox IP through the ID 6. To see Modbus read addresses, refer to section Modbus memory map registers [► 13].

Read example

- Modbus port: 502
- Device Address: 6
- Function Code: 03
- Register Address: 48695
- Number of registers: 2

Sensor Signal Converters 1-3 Hail Sensors 1-2

To monitor Sensor Signal Converters 1 Hail Sensor 2 in Block 6, SCADA must be connected to the TBox IP through the ID 6. To see Modbus read addresses, refer to section Modbus memory map registers [► 13].

Read example

- Modbus port: 502
- Device Address: 6
- Function Code: 03
- Register Address: 48695
- Number of registers: 2

Sensor Signal Converters 1-3 Snow Sensors 1-2

To monitor Sensor Signal Converters 1 Hail Sensor 2 in Block 6, SCADA must be connected to the TBox IP through the ID 6. To see Modbus read addresses, refer to section Modbus memory map registers [► 13].

Read example

- Modbus port: 502
- Device Address: 6
- Function Code: 03
- Register Address: 48695
- Number of registers: 2

6.3.3. DBox status description

This section introduces various status labels derived from the reported operation mode, alarms of each tracker, and TBox command variables.

NOTICE



Table values explanation

Alarms are the most significant values that must be considered to determine the Status Label, having more importance compared to the rest of the variables.

a) The dash (-) indicates that the value of the field is less significant compared to other variables identified as priorities.

b) The labelled **NO** means that it cannot generate any active alarms.

System version 3.2

OPERATION MODE CODE	OPERATION MODE	ALARM BIT	ALARMS	TBOX VARIABLE	POSITION CODE VALUE	POSITION CODE	STATUS LABEL	DESCRIPTION
0	Defect: Not recognized	NO	NO	-	-	-	Defect: Not recognized	TBox still does not have any information about DBox.
1	TBox Control: Auto	NO	NO	-	0	Tracking	TBox Control: Auto	DBox is receiving the solar setpoint and tracking the sun as usual.

OPERATION MODE CODE	OPERATION MODE	ALARM BIT	ALARMS	TBOX VARIABLE	POSITION CODE VALUE	POSITION CODE	STATUS LABEL	DESCRIPTION
1	TBox Control: Auto	NO	NO	WindAlarm & Snowstate & Diffusestate & Hailstate = FALSE AND Backtrackingstatus = TRUE	2	Backtracki ng	TBox Control: Backtracking	TBox is sending backtracking setpoints.
1	TBox Control: Auto	NO	NO	WindAlarm & Snowstate & Diffusestate & Hailstate = FALSE AND Nightstatus = TRUE	1	Night position	TBox Control: Night Mode	TBox is sending backtracking setpoints.
1	TBox Control: Auto	NO	NO	WindAlarm & Snowstate & Hailstate = FALSE AND Diffusestate = TRUE	3	Diffuse	TBox Control: Diffuse Mode Active	TBox is sending backtracking setpoints.

OPERATION MODE CODE	OPERATION MODE	ALARM BIT	ALARMS	TBOX VARIABLE	POSITION CODE VALUE	POSITION CODE	STATUS LABEL	DESCRIPTION
1	TBox Control: Auto	NO	NO	WindAlarm = TRUE AND ElevationPosition sign = GlobalWindDirectio n (East negative, West Positive) AND Abs(ElevationPositi on) ~ 15 or 30	5	Wind Defense	TBox Control: Auto Wind Stow	The Wind Alarm has triggered and DBox is receiving it normally.
				WindAlarm = FALSE AND Snowstate = TRUE	7	Snow position	TBox Control: Auto Snow Alarm	TBox is sending backtracking setpoints.
				WindAlarm & Snowstate = FALSE AND Hailstate = TRUE	8	Hail position	TBox Control: Hail Alarm Active	TBox is sending backtracking setpoints.

OPERATION MODE CODE	OPERATION MODE	ALARM BIT	ALARMS	TBOX VARIABLE	POSITION CODE VALUE	POSITION CODE	STATUS LABEL	DESCRIPTION
1	TBox Control: Auto	2	Inclinomet er fault	-	-	-	Defect: Inclinometer fault	An error in the inclinometer has occurred. It is necessary to check DBox locally.
2	Remote Control: Manual Position	2	Inclinomet er fault	-	-	-	Defect: Inclinometer fault	An error in the inclinometer has occurred. It is necessary to check DBox locally.
3	Local Control: PVH Application	2	Inclinomet er fault	-	-	-	Defect: Inclinometer fault	An error in the inclinometer has occurred. It is necessary to check DBox locally.
4	Halt: Waiting for acknowledge	2	Inclinomet er fault	-	-	-	Defect: Inclinometer fault	An error in the inclinometer has occurred. It is necessary to check DBox locally.

OPERATION MODE CODE	OPERATION MODE	ALARM BIT	ALARMS	TBOX VARIABLE	POSITION CODE VALUE	POSITION CODE	STATUS LABEL	DESCRIPTION
5	Remote Control: Manual Wind Position	2	Inclinomet er fault	-	-	-	Defect: Inclinometer fault	An error in the inclinometer has occurred. It is necessary to check DBox locally.
10	Defect: TBox communicati on lost	2	Inclinomet er fault	-	-	-	Defect: Inclinometer fault	An error in the inclinometer has occurred. It is necessary to check DBox locally.
11	Local Control: Buttons	2	Inclinomet er fault	-	-	-	Defect: Inclinometer fault	An error in the inclinometer has occurred. It is necessary to check DBox locally.
12	Remote Control: Manual Construction Position	2	Inclinomet er fault	-	-	-	Defect: Inclinometer fault	An error in the inclinometer has occurred. It is necessary to check DBox locally.

OPERATION MODE CODE	OPERATION MODE	ALARM BIT	ALARMS	TBOX VARIABLE	POSITION CODE VALUE	POSITION CODE	STATUS LABEL	DESCRIPTION
1	TBox Control: Auto	4	Overcurrent fault	-	-	-	Defect: Overcurrent fault	An overcurrent in the output of DBox has happened. The structure may not be correctly mounted or there may be something overloading. It is necessary to go locally to check DBox.
2	Remote Control: Manual Position	4	Overcurrent fault	-	-	-	Defect: Overcurrent fault	An overcurrent in the output of DBox has happened. The structure may not be correctly mounted or there may be something overloading. It is necessary to go locally to check DBox.

OPERATION MODE CODE	OPERATION MODE	ALARM BIT	ALARMS	TBOX VARIABLE	POSITION CODE VALUE	POSITION CODE	STATUS LABEL	DESCRIPTION
3	Local Control: PVH Application	4	Overcurrent fault	-	-	-	Defect: Overcurrent fault	An overcurrent in the output of DBox has happened. The structure may not be correctly mounted or there may be something overloading. It is necessary to go locally to check DBox.
4	Halt: Waiting for acknowledge	4	Overcurrent fault	-	-	-	Defect: Overcurrent fault	An overcurrent in the output of DBox has happened. The structure may not be correctly mounted or there may be something overloading. It is necessary to go locally to check DBox.

OPERATION MODE CODE	OPERATION MODE	ALARM BIT	ALARMS	TBOX VARIABLE	POSITION CODE VALUE	POSITION CODE	STATUS LABEL	DESCRIPTION
5	Remote Control: Manual Wind Position	4	Overcurre nt fault	-	-	-	Defect: Overcurrent fault	An overcurrent in the output of DBox has happened. The structure may not be correctly mounted or there may be something overloading. It is necessary to go locally to check DBox.
10	Defect: TBox communicati on lost	4	Overcurre nt fault	-	-	-	Defect: Overcurrent fault	An overcurrent in the output of DBox has happened. The structure may not be correctly mounted or there may be something overloading. It is necessary to go locally to check DBox.

OPERATION MODE CODE	OPERATION MODE	ALARM BIT	ALARMS	TBOX VARIABLE	POSITION CODE VALUE	POSITION CODE	STATUS LABEL	DESCRIPTION
11	Local Control: Buttons	4	Overcurrent fault	-	-	-	Defect: Overcurrent fault	An overcurrent in the output of DBox has happened. The structure may not be correctly mounted or there may be something overloading. It is necessary to go locally to check DBox.
12	Remote Control: Manual Construction Position	4	Overcurrent fault	-	-	-	Defect: Overcurrent fault	An overcurrent in the output of DBox has happened. The structure may not be correctly mounted or there may be something overloading. It is necessary to go locally to check DBox.

OPERATION MODE CODE	OPERATION MODE	ALARM BIT	ALARMS	TBOX VARIABLE	POSITION CODE VALUE	POSITION CODE	STATUS LABEL	DESCRIPTION
-	-	-	-	Notification (DBox): Bit 0. Outdated DBox Data	-	-	Defect: Outdated DBox Data	TBox has not been receiving updated information about the referred DBox for more than an hour. It is necessary to go locally to check if DBox is powered.
-	-	2 or more Bits active	2 or more Alarms active	-	-	-	Defect: Several faults	DBox is notifying of several faults. Check them one by one.
2	Remote Control: Manual Position	NO	NO	-	9	User position	Remote Control: Manual Position	DBox is being commanded in remote mode with a manual movement.
3	Local Control: PVH Application	NO	NO	-	9	User position	Local Control: PVH Application	DBox is being commanded in local mode: from the NFC APP for smartphone.

OPERATION MODE CODE	OPERATION MODE	ALARM BIT	ALARMS	TBOX VARIABLE	POSITION CODE VALUE	POSITION CODE	STATUS LABEL	DESCRIPTION
4	Halt: Waiting for acknowledge	NO	NO	-	-	-	Halt: Waiting for acknowledge	DBox is halted waiting for any manual intervention. It may be local (touch a button or local control or set an NFC command from NFC APP) or remote (send "SetAutoControlFl ag" from TBox).
5	Remote Control: Manual Wind Position	NO	NO	-	9	User position	Remote Control: Manual Wind Position	DBox is set in Stow Position commanded manually in remote mode.
6	Defect: Secure Battery	NO	NO	-	-	-	Defect: Secure Battery	When the battery SoC is under a threshold, DBox stops its movement but remains powered to transmit the information to TBox. Wait for it to charge.

OPERATION MODE CODE	OPERATION MODE	ALARM BIT	ALARMS	TBOX VARIABLE	POSITION CODE VALUE	POSITION CODE	STATUS LABEL	DESCRIPTION
6	Defect: Secure Battery	1	Charger Fault	-	-	-	Defect: Solar Charger fault	There is an error in the solar charger information. When the charger fails, DBox cannot read any level of battery, so it assumes that the charge level is 0% and takes the structure to stow position. It is necessary to locally check DBox.
7	Defect: Fault Stop	3	Motor Wrong Direction Fault	-	-	-	Defect: fault Stop	Wrong motor behavior has occurred. It is necessary to locally check DBox. When the alarm is cleaned, reset DBox.
8	Local Control: Emergency Stop	0	Emergenc y Stop Button Pressed	-	10	Stop position	Local Control: Emergency Stop	DBox emergency stop button has been pressed.

OPERATION MODE CODE	OPERATION MODE	ALARM BIT	ALARMS	TBOX VARIABLE	POSITION CODE VALUE	POSITION CODE	STATUS LABEL	DESCRIPTION
9	Remote Control: Manual Stop	NO	NO	-	10	Stop position	Remote Control: Manual Stop	A remote emergency stop is being sent from TBox to DBox.
10	Defect: TBox communicati on lost	NO	NO	-	-	-	Defect: TBox communicati on lost	DBox is reporting that the timeout without receiving information from TBox has expired and it is in a "No communications" status until it receives a new message.
11	Local Control: Buttons	NO	NO	-	9	User position	Local Control: Buttons	DBox is being commanded in a local mode: the capacitive buttons on the cover of DBox are being pressed.
12	Remote Control: Manual Construction Position	NO	NO	-	9	User position	Remote Control: Manual Construction Position	DBox is set in Construction Position commanded manually in remote mode.

System version 3.1

OPERATION MODE CODE	OPERATION MODE	ALARM BIT	ALARMS	TBOX VARIABLE	STATUS LABEL	DESCRIPTION
0	Defect: Not recognized	NO	NO	-	Defect: Not recognized	TBox still does not have any information about DBox.
1	TBox Control: Auto	NO	NO	-	TBox Control: Auto	DBox is receiving the solar setpoint and tracking the sun as usual.
1	TBox Control: Auto	NO	NO	WindAlarm & Snowstate & Diffusestate & Hailstate = FALSE AND Backtrackingstatus = TRUE	TBox Control: Backtracking	TBox is sending backtracking setpoints.
1	TBox Control: Auto	NO	NO	WindAlarm & Snowstate & Diffusestate & Hailstate = FALSE AND Nightstatus = TRUE	TBox Control: Night Mode	TBox is sending backtracking setpoints.
1	TBox Control: Auto	NO	NO	WindAlarm & Snowstate & Hailstate = FALSE AND Diffusestate = TRUE	TBox Control: Diffuse Mode Active	TBox is sending backtracking setpoints.

OPERATION MODE CODE	OPERATION MODE	ALARM BIT	ALARMS	TBOX VARIABLE	STATUS LABEL	DESCRIPTION
1	TBox Control: Auto	NO	NO	WindAlarm = TRUE	TBox Control: Auto Wind Stow	The Wind Alarm has triggered and DBox is receiving it normally.
				AND		
				ElevationPosition sign = GlobalWindDirection (East negative, West Positive)		
				AND		
1	TBox Control: Auto	NO	NO	Abs (ElevationPosition) ~ 15 or 30	TBox Control: Auto Snow Alarm	TBox is sending backtracking setpoints.
				WindAlarm = FALSE		
				AND		
				Snowstate = TRUE		
1	TBox Control: Auto	NO	NO	WindAlarm & Snowstate = FALSE	TBox Control: Hail Alarm Active	TBox is sending backtracking setpoints.
				AND		
				Hailstate = TRUE		
1	TBox Control: Auto	2	Inclinometer fault	-	Defect: Inclinometer fault	An error in the inclinometer has occurred. It is necessary to check DBox locally.
2	Remote Control: Manual Position	2	Inclinometer fault	-	Defect: Inclinometer fault	An error in the inclinometer has occurred. It is necessary to check DBox locally.

OPERATION MODE CODE	OPERATION MODE	ALARM BIT	ALARMS	TBOX VARIABLE	STATUS LABEL	DESCRIPTION
3	Local Control: PVH Application	2	Inclinometer fault	-	Defect: Inclinometer fault	An error in the inclinometer has occurred. It is necessary to check DBox locally.
4	Halt: Waiting for acknowledge	2	Inclinometer fault	-	Defect: Inclinometer fault	An error in the inclinometer has occurred. It is necessary to check DBox locally.
5	Remote Control: Manual Wind Position	2	Inclinometer fault	-	Defect: Inclinometer fault	An error in the inclinometer has occurred. It is necessary to check DBox locally.
10	Defect: TBox communication lost	2	Inclinometer fault	-	Defect: Inclinometer fault	An error in the inclinometer has occurred. It is necessary to check DBox locally.
11	Local Control: Buttons	2	Inclinometer fault	-	Defect: Inclinometer fault	An error in the inclinometer has occurred. It is necessary to check DBox locally.
12	Remote Control: Manual Construction Position	2	Inclinometer fault	-	Defect: Inclinometer fault	An error in the inclinometer has occurred. It is necessary to check DBox locally.
1	TBox Control: Auto	4	Overcurrent fault	-	Defect: Overcurrent fault	An overcurrent in the output of DBox has happened. The structure may not be correctly mounted or there may be something overloading. It is necessary to go locally to check DBox.

OPERATION MODE CODE	OPERATION MODE	ALARM BIT	ALARMS	TBOX VARIABLE	STATUS LABEL	DESCRIPTION
2	Remote Control: Manual Position	4	Overcurrent fault	-	Defect: Overcurrent fault	An overcurrent in the output of DBox has happened. The structure may not be correctly mounted or there may be something overloading. It is necessary to go locally to check DBox.
3	Local Control: PVH Application	4	Overcurrent fault	-	Defect: Overcurrent fault	An overcurrent in the output of DBox has happened. The structure may not be correctly mounted or there may be something overloading. It is necessary to go locally to check DBox.
4	Halt: Waiting for acknowledge	4	Overcurrent fault	-	Defect: Overcurrent fault	An overcurrent in the output of DBox has happened. The structure may not be correctly mounted or there may be something overloading. It is necessary to go locally to check DBox.
5	Remote Control: Manual Wind Position	4	Overcurrent fault	-	Defect: Overcurrent fault	An overcurrent in the output of DBox has happened. The structure may not be correctly mounted or there may be something overloading. It is necessary to go locally to check DBox.

OPERATION MODE CODE	OPERATION MODE	ALARM BIT	ALARMS	TBOX VARIABLE	STATUS LABEL	DESCRIPTION
10	Defect: TBox communication lost	4	Overcurrent fault	-	Defect: Overcurrent fault	An overcurrent in the output of DBox has happened. The structure may not be correctly mounted or there may be something overloading. It is necessary to go locally to check DBox.
11	Local Control: Buttons	4	Overcurrent fault	-	Defect: Overcurrent fault	An overcurrent in the output of DBox has happened. The structure may not be correctly mounted or there may be something overloading. It is necessary to go locally to check DBox.
12	Remote Control: Manual Construction Position	4	Overcurrent fault	-	Defect: Overcurrent fault	An overcurrent in the output of DBox has happened. The structure may not be correctly mounted or there may be something overloading. It is necessary to go locally to check DBox.
-	-	15	Outdated DBox Data	-	Defect: Outdated DBox Data	TBox has not been receiving updated information about the referred DBox for more than an hour. It is necessary to go locally to check if DBox is powered.

OPERATION MODE CODE	OPERATION MODE	ALARM BIT	ALARMS	TBOX VARIABLE	STATUS LABEL	DESCRIPTION
-	-	2 or more Bits active	2 or more Alarms active	-	Defect: Several Faults	DBox is notifying of several faults. Check them one by one.
2	Remote Control: Manual Position	NO	NO	-	Remote Control: Manual Position	DBox is being commanded in remote mode with a manual movement.
3	Local Control: PVH Application	NO	NO	-	Local Control: PVH Application	DBox is being commanded in local mode: from the NFC APP for smartphone.
4	Halt: Waiting for acknowledge	NO	NO	-	Halt: Waiting for acknowledge	DBox is halted waiting for any manual intervention. It may be local (touch a button or local control or set an NFC command from NFC APP) or remote (send "SetAutoControlFlag" from TBox)
5	Remote Control: Manual Wind Position	NO	NO	-	Remote Control: Manual Wind Position	DBox is set in Stow Position commanded manually in remote mode.
6	Defect: Secure Battery	NO	NO	-	Defect: Secure Battery	When the battery SoC is under a threshold, DBox stops its movement but remains powered to transmit the information to TBox. Wait for it to charge.

OPERATION MODE CODE	OPERATION MODE	ALARM BIT	ALARMS	TBOX VARIABLE	STATUS LABEL	DESCRIPTION
6	Defect: Secure Battery	1	Charger Fault	-	Defect: Solar Charger Fault	There is an error in the solar charger information. When the charger fails, DBox cannot read any level of battery, so it assumes that the charge level is 0% and takes the structure to stow position. It is necessary to locally check DBox.
7	Defect: Fault Stop	3	Motor Wrong Direction Fault	-	Defect: Fault Stop	Wrong motor behavior has occurred. It is necessary to locally check DBox. When the alarm is cleaned, reset DBox.
8	Local Control: Emergency Stop	0	Emergency Stop Button Pressed	-	Local Control: Emergency Stop	DBox emergency stop button has been pressed.
9	Remote Control: Manual Stop	NO	NO	-	Remote Control: Manual Stop	A remote emergency stop is being sent from TBox to DBox.
10	Defect: TBox communication lost	NO	NO	-	Defect: TBox communication lost	DBox is reporting that the timeout without receiving information from TBox has expired and it is in a "No communications" status until it receives a new message.

OPERATION MODE CODE	OPERATION MODE	ALARM BIT	ALARMS	TBOX VARIABLE	STATUS LABEL	DESCRIPTION
11	Local Control: Buttons	NO	NO	-	Local Control: Buttons	DBox is being commanded in a local mode: the capacitive buttons on the cover of DBox are being pressed.
12	Remote Control: Manual Construction Position	NO	NO	-	Remote Control: Manual Construction Position	DBox is set in Construction Position commanded manually in remote mode.

6.3.4. Meteo status description

The following table applies to both system version 3.1 and system version 3.2.

ALARM BIT	ALARMS	VARIABLE	STATUS LABEL	DESCRIPTION
NO	NO	-	Nominal	Meteo is working in nominal conditions.
0	Anemometer fault	-	Anemometer fault	Meteo is not able to read the anemometer information.
1	Charger fault	BatteryLevel = 0	Charger fault	The level of battery is read from the solar charger. When the charger fails, Meteo cannot read any level of battery, so it assumes the charge level is 0 %.

7. Server Port 503 - Modbus Memory Map

7.1. Modbus map overview

This section describes the information available on Modbus port 503. To retrieve this information, connect a Modbus client to the port.

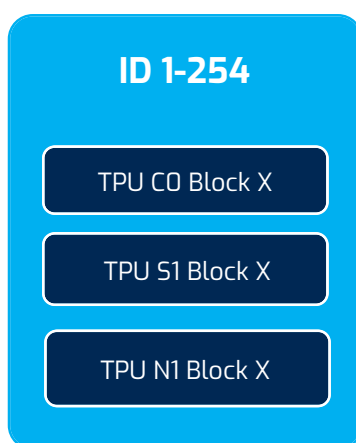


Figure 3. Modbus port 503

The number of the desired block to be connected corresponds to the Modbus ID used for the connection. Each block is configured to display one Meteo entry and up to 500 DBox entries, which are ordered consecutively.

TPU refers to devices controlled by trackers. Each TPU operates a motor and is driven by DBox.

7.2. Modbus memory map registers

7.2.1. Memory map distribution

TAG	MODBUS ID	FIRST ADDRESS	LAST ADDRESS	DESCRIPTION
TPU C0 1-500	1-254	40000	44999	INFORMATION OF CENTRAL TPU
TPU S1 1-500	1-254	45000	49999	INFORMATION OF SOUTH TPU
TPU N1 1-500	1-254	50000	54999	INFORMATION OF NORTH TPU

7.2.2. ID 1 to 254 - Power Block Map

TPU C0 1										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	State	40000	0	Uint	-	0	65535	-	R	State: - Value 0. Preproduction: A configuration has not been received yet - Value 1. Halt: Waiting for acknowledge to start working - Value 2. Defect: One or more alarms have been detected, movement is stopped - Value 3. Remote Control: Tracking - Value 4. Temperature Exception: Temperature limits are exceeded - Value 255. Non-Existing: This TPU slot is not used
	ElevationSetpoint	40001	1	Int	Degrees	-7000	7000	100	R	Target elevation in degrees from horizontal. Auto or manual depending on mode of operation

TPU CO 1										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	ElevationPosition	40002	2	Int	Deg ree s	-7000	7000	100	R	Actual elevation position in degrees from horizontal
Alarms:										
FEEDBACK	Alarms	40003	3	Uint	-	0	65535	-	R	• Bit 0. Inclinator Fault • Bit 1. Wrong direction movement • Bit 2. Overcurrent Fault • Bit 3. Slow motor movement • Bit 4. High power consumption • Bit 5. Out of boundaries • Bit 6. TPU Communication Error • Bit 7. Low temperature • Bit 8. High temperature
FEEDBACK	MaxMotorCurrent	40004	4	Uint	mA	0	2000 0	1	R	Maximum motor current of the last performed movement without considering the distance between its first and last angle

TPU CO 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	State	40010	10	Uint	-	0	65535	-	R	State:
										• Value 0. Preproduction: A configuration has not been received yet
										• Value 1. Halt: Waiting for acknowledge to start working
										• Value 2. Defect: One or more alarms have been detected, movement is stopped
										• Value 3. Remote Control: Tracking
										• Value 4. Temperature Exception: Temperature limits are exceeded
										• Value 255. Non-Existing: This TPU slot is not used
FEEDBACK	ElevationSetpoint	40011	11	Int	Degrees	-7000	7000	100	R	Target elevation in degrees from horizontal. Auto or manual depending on mode of operation
FEEDBACK	ElevationPosition	40012	12	Int	Degrees	-7000	7000	100	R	Actual elevation position in degrees from horizontal

TPU CO 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Alarms	40013	13	Uint	-	0	65535	-	R	Alarms:
										• Bit 0. Inclinometer Fault • Bit 1. Wrong direction movement • Bit 2. Overcurrent Fault • Bit 3. Slow motor movement • Bit 4. High power consumption • Bit 5. Out of boundaries • Bit 6. TPU Communication Error • Bit 7. Low temperature • Bit 8. High temperature
FEEDBACK	MaxMotorCurrent	40014	14	Uint	mA	0	20000	1	R	Maximum motor current of the last performed movement without considering the distance between its first and last angle

TPU CO 3										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	State	40020	20	Uint	-	0	65535	-	R	State:
										• Value 0. Preproduction: A configuration has not been received yet
										• Value 1. Halt: Waiting for acknowledge to start working
										• Value 2. Defect: One or more alarms have been detected, movement is stopped
										• Value 3. Remote Control: Tracking
										• Value 4. Temperature Exception: Temperature limits are exceeded
										• Value 255. Non-Existing: This TPU slot is not used
FEEDBACK	ElevationSetpoint	40021	21	Int	Degrees	-7000	7000	100	R	Target elevation in degrees from horizontal. Auto or manual depending on mode of operation
FEEDBACK	ElevationPosition	40022	22	Int	Degrees	-7000	7000	100	R	Actual elevation position in degrees from horizontal

TPU CO 3										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Alarms	40023	23	Uint	-	0	65535	-	R	Alarms:
										• Bit 0. Inclinomater Fault • Bit 1. Wrong direction movement • Bit 2. Overcurrent Fault • Bit 3. Slow motor movement • Bit 4. High power consumption • Bit 5. Out of boundaries • Bit 6. TPU Communication Error • Bit 7. Low temperature • Bit 8. High temperature
FEEDBACK	MaxMotorCurrent	40024	24	Uint	mA	0	20000	1	R	Maximum motor current of the last performed movement without considering the distance between its first and last angle

TPU CO 4-500										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	State	...	...	Uint	-	0	65535	-	R	State: - Value 0. Preproduction: A configuration has not been received yet - Value 1. Halt: Waiting for acknowledge to start working - Value 2. Defect: One or more alarms have been detected, movement is stopped - Value 3. Remote Control: Tracking - Value 4. Temperature Exception: Temperature limits are exceeded - Value 255. Non-Existing: This TPU slot is not used
										Target elevation in degrees from horizontal. Auto or manual depending on mode of operation
FEEDBACK	ElevationPosition	...	...	Int	Degrees	-7000	7000	100	R	Actual elevation position in degrees from horizontal

TPU CO 4-500										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Alarms	...	...	Uint	-	0	65535	-	R	Alarms:
										• Bit 0. Inclinator Fault • Bit 1. Wrong direction movement • Bit 2. Overcurrent Fault • Bit 3. Slow motor movement • Bit 4. High power consumption • Bit 5. Out of boundaries • Bit 6. TPU Communication Error • Bit 7. Low temperature • Bit 8. High temperature
FEEDBACK	MaxMotorCurrent	...	...	Uint	mA	0	20000	1	R	Maximum motor current of the last performed movement without considering the distance between its first and last angle

TPU S1 1										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	State	45000	5000	Uint	-	0	65535	-	R	State:
										• Value 0. Preproduction: A configuration has not been received yet
										• Value 1. Halt: Waiting for acknowledge to start working
										• Value 2. Defect: One or more alarms have been detected, movement is stopped
										• Value 3. Remote Control: Tracking
FEEDBACK	ElevationSetpoint	45001	5001	Int	Degrees	-7000	7000	100	R	• Value 4. Temperature Exception: Temperature limits are exceeded
										• Value 255. Non-Existing: This TPU slot is not used
										Target elevation in degrees from horizontal. Auto or manual depending on mode of operation
										Actual elevation position in degrees from horizontal

TPU S1 1										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Alarms	45003	5003	Uint	-	0	65535	-	R	Alarms:
										• Bit 0. Inclinator Fault • Bit 1. Wrong direction movement • Bit 2. Overcurrent Fault • Bit 3. Slow motor movement • Bit 4. High power consumption • Bit 5. Out of boundaries • Bit 6. TPU Communication Error • Bit 7. Low temperature • Bit 8. High temperature
FEEDBACK	MaxMotorCurrent	45004	5004	Uint	mA	0	20000	1	R	Maximum motor current of the last performed movement without considering the distance between its first and last angle

TPU S1 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	State	45010	5010	Uint	-	0	65535	-	R	State: - Value 0. Preproduction: A configuration has not been received yet - Value 1. Halt: Waiting for acknowledge to start working - Value 2. Defect: One or more alarms have been detected, movement is stopped - Value 3. Remote Control: Tracking - Value 4. Temperature Exception: Temperature limits are exceeded - Value 255. Non-Existing: This TPU slot is not used
	ElevationSetpoint	45011	5011	Int	Degrees	-7000	7000	100	R	Target elevation in degrees from horizontal. Auto or manual depending on mode of operation
	ElevationPosition	45012	5012	Int	Degrees	-7000	7000	100	R	Actual elevation position in degrees from horizontal

TPU S1 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Alarms	45013	5013	Uint	-	0	65535	-	R	Alarms:
										• Bit 0. Inclinator Fault • Bit 1. Wrong direction movement • Bit 2. Overcurrent Fault • Bit 3. Slow motor movement • Bit 4. High power consumption • Bit 5. Out of boundaries • Bit 6. TPU Communication Error • Bit 7. Low temperature • Bit 8. High temperature
FEEDBACK	MaxMotorCurrent	45014	5014	Uint	mA	0	20000	1	R	Maximum motor current of the last performed movement without considering the distance between its first and last angle

TPU S13										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	State	45020	5020	Uint	-	0	65535	-	R	State: - Value 0. Preproduction: A configuration has not been received yet - Value 1. Halt: Waiting for acknowledge to start working - Value 2. Defect: One or more alarms have been detected, movement is stopped - Value 3. Remote Control: Tracking - Value 4. Temperature Exception: Temperature limits are exceeded - Value 255. Non-Existing: This TPU slot is not used
										Target elevation in degrees from horizontal. Auto or manual depending on mode of operation
										Actual elevation position in degrees from horizontal

TPU S13										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Alarms	45023	5023	Uint	-	0	65535	-	R	Alarms: • Bit 0. Inclinator Fault • Bit 1. Wrong direction movement • Bit 2. Overcurrent Fault • Bit 3. Slow motor movement • Bit 4. High power consumption • Bit 5. Out of boundaries • Bit 6. TPU Communication Error • Bit 7. Low temperature • Bit 8. High temperature
	MaxMotorCurrent	45024	5024	Uint	mA	0	20000	1	R	Maximum motor current of the last performed movement without considering the distance between its first and last angle

TPU S1 4-500										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	State	...	...	Uint	-	0	65535	-	R	State: - Value 0. Preproduction: A configuration has not been received yet - Value 1. Halt: Waiting for acknowledge to start working - Value 2. Defect: One or more alarms have been detected, movement is stopped - Value 3. Remote Control: Tracking - Value 4. Temperature Exception: Temperature limits are exceeded - Value 255. Non-Existing: This TPU slot is not used
	ElevationSetpoint	...	...	Int	Degrees	-7000	7000	100	R	Target elevation in degrees from horizontal. Auto or manual depending on mode of operation
	ElevationPosition	...	...	Int	Degrees	-7000	7000	100	R	Actual elevation position in degrees from horizontal

TPU S1 4-500										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Alarms	...	...	Uint	-	0	65535	-	R	Alarms:
										• Bit 0. Inclinator Fault • Bit 1. Wrong direction movement • Bit 2. Overcurrent Fault • Bit 3. Slow motor movement • Bit 4. High power consumption • Bit 5. Out of boundaries • Bit 6. TPU Communication Error • Bit 7. Low temperature • Bit 8. High temperature
FEEDBACK	MaxMotorCurrent	...	...	Uint	mA	0	20000	1	R	Maximum motor current of the last performed movement without considering the distance between its first and last angle

TPU N1 1										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	State	50000	10000	Uint	-	0	65535	-	R	State:
										• Value 0. Preproduction: A configuration has not been received yet
										• Value 1. Halt: Waiting for acknowledge to start working
										• Value 2. Defect: One or more alarms have been detected, movement is stopped
										• Value 3. Remote Control: Tracking
										• Value 4. Temperature Exception: Temperature limits are exceeded
										• Value 255. Non-Existing: This TPU slot is not used
FEEDBACK	ElevationSetpoint	50001	10001	Int	Degrees	-7000	7000	100	R	Target elevation in degrees from horizontal. Auto or manual depending on mode of operation
FEEDBACK	ElevationPosition	50002	10002	Int	Degrees	-7000	7000	100	R	Actual elevation position in degrees from horizontal

TPU N1 1										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Alarms	50003	10003	Uint	-	0	65535	-	R	Alarms: • Bit 0. Inclinator Fault • Bit 1. Wrong direction movement • Bit 2. Overcurrent Fault • Bit 3. Slow motor movement • Bit 4. High power consumption • Bit 5. Out of boundaries • Bit 6. TPU Communication Error • Bit 7. Low temperature • Bit 8. High temperature
	MaxMotorCurrent	50004	10004	Uint	mA	0	20000	1	R	Maximum motor current of the last performed movement without considering the distance between its first and last angle

TPU N1 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	State	50010	10010	Uint	-	0	65535	-	R	State: - Value 0. Preproduction: A configuration has not been received yet - Value 1. Halt: Waiting for acknowledge to start working - Value 2. Defect: One or more alarms have been detected, movement is stopped - Value 3. Remote Control: Tracking - Value 4. Temperature Exception: Temperature limits are exceeded - Value 255. Non-Existing: This TPU slot is not used
	ElevationSetpoint	50011	10011	Int	Degrees	-7000	7000	100	R	Target elevation in degrees from horizontal. Auto or manual depending on mode of operation
	ElevationPosition	50012	10012	Int	Degrees	-7000	7000	100	R	Actual elevation position in degrees from horizontal

TPU N1 2										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Alarms	50013	10013	Uint	-	0	65535	-	R	Alarms:
										• Bit 0. Inclinometer Fault • Bit 1. Wrong direction movement • Bit 2. Overcurrent Fault • Bit 3. Slow motor movement • Bit 4. High power consumption • Bit 5. Out of boundaries • Bit 6. TPU Communication Error • Bit 7. Low temperature • Bit 8. High temperature
FEEDBACK	MaxMotorCurrent	50014	10014	Uint	mA	0	20000	1	R	Maximum motor current of the last performed movement without considering the distance between its first and last angle

TPU N1 3										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	State	50020	10020	Uint	-	0	65535	-	R	State: - Value 0. Preproduction: A configuration has not been received yet - Value 1. Halt: Waiting for acknowledge to start working - Value 2. Defect: One or more alarms have been detected, movement is stopped - Value 3. Remote Control: Tracking - Value 4. Temperature Exception: Temperature limits are exceeded - Value 255. Non-Existing: This TPU slot is not used
										Target elevation in degrees from horizontal. Auto or manual depending on mode of operation
										Actual elevation position in degrees from horizontal

TPU N1 3										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Alarms	50023	10023	Uint	-	0	65535	-	R	Alarms:
										• Bit 0. Inclinometer Fault • Bit 1. Wrong direction movement • Bit 2. Overcurrent Fault • Bit 3. Slow motor movement • Bit 4. High power consumption • Bit 5. Out of boundaries • Bit 6. TPU Communication Error • Bit 7. Low temperature • Bit 8. High temperature
FEEDBACK	MaxMotorCurrent	50024	10024	Uint	mA	0	20000	1	R	Maximum motor current of the last performed movement without considering the distance between its first and last angle

TPU N1 4-500										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	State	...	...	Uint	-	0	65535	-	R	State:
										• Value 0. Preproduction: A configuration has not been received yet
										• Value 1. Halt: Waiting for acknowledge to start working
										• Value 2. Defect: One or more alarms have been detected, movement is stopped
										• Value 3. Remote Control: Tracking
FEEDBACK	ElevationSetpoint	...	...	Int	Degrees	-7000	7000	100	R	• Value 4. Temperature Exception: Temperature limits are exceeded
										• Value 255. Non-Existing: This TPU slot is not used
										Target elevation in degrees from horizontal. Auto or manual depending on mode of operation
										Actual elevation position in degrees from horizontal

TPU N1 4-500										
REGISTER TYPE	TagID	ADDRESS	HOLDING REGISTER	DATA TYPE	UNITS	MIN	MAX	SCALE FACTOR	ACCESS	DESCRIPTION
FEEDBACK	Alarms	...	...	Uint	-	0	65535	-	R	Alarms:
										• Bit 0. Inclinator Fault • Bit 1. Wrong direction movement • Bit 2. Overcurrent Fault • Bit 3. Slow motor movement • Bit 4. High power consumption • Bit 5. Out of boundaries • Bit 6. TPU Communication Error • Bit 7. Low temperature • Bit 8. High temperature
FEEDBACK	MaxMotorCurrent	...	...	Uint	mA	0	20000	1	R	Maximum motor current of the last performed movement without considering the distance between its first and last angle

7.3. Integration examples

TPU CO 1-500

To monitor TPU CO 166 in Block 23, SCADA must be connected to the TBox IP through the ID 23. Refer to Modbus read addresses, refer to section Modbus memory map registers (► 117).

Read example

- Modbus port: 503
- Device Address: 23
- Function Code: 03
- Register Address: 41650
- Number of registers: 5

TPU S1 1-500

To monitor TPU S1 93 in Block 17, SCADA must be connected to the TBox IP through the ID 17. Refer to Modbus read addresses, refer to section Modbus Memory Map registers (► 117).

Read example

- Modbus port: 503
- Device Address: 17
- Function Code: 03
- Register Address: 45920
- Number of registers: 5

TPU N1 1-500

To monitor TPU N1 403 in Block 5, SCADA must be connected to the TBox IP through the ID 5. Refer to Modbus read addresses, refer to section Modbus Memory Map registers (► 117).

Read example

- Modbus port: 503
- Device Address: 5
- Function Code: 03
- Register Address: 54090
- Number of registers: 5

8. PVH Support

Any doubt regarding the PVH solar tracker control system or the requirements prior to the installation of the PVH solar tracker control system can be asked to the Presales team or the Aftersales team, depending on the phase of the project.



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